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The adaptive Gaussian mixtures unscented Kalman filter for attitude determination using light curves

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Abstract

The Adaptive Gaussian Mixtures Unscented Kalman Filter (AGMUKF) is introduced to estimate the attitude of a Resident Space Object using light curves. This filter models the state probability density function as a Gaussian Mixture. This enables to capture the non-linearities of the light-curve measurement model. A non-linearity index is used to refine the mixture when necessary, and individual Gaussian kernels are merged back together when their relative distance is below a certain threshold. A conventional attitude Unscented Kalman Filter (UKF) is used to propagate and update each kernel. The AGMUKF efficiently maintains the mixture population as low as possible, while still being able to represent non-symmetric, multimodal, arbitrarily complex distributions. Therefore, it is presented as a promising alternative to Particle-Filter-based implementations, the current state of the art used in sequential attitude estimation from light curves. The non-linearity index has been used to show that the measurement model is the main contributor to the system non-linearity. Results have demonstrated the superiority of the AGMUKF w.r.t. the UKF for attitude determination, and that it can converge for high initial state uncertainty cases, successfully capturing the non-Gaussian probability distribution of the state.

Keywords: Object characterization; Gaussian mixtures filter; Unscented Kalman filter; Attitude determination; Light curve

1. Introduction

Within the sector of Space Surveillance and Tracking (SST), Attitude Determination (AD) of Resident Space Objects (RSOs) is key to guarantee the safe and feasible usage of the Earth's space environment. Potential uses include status monitoring, precise orbit propagation and providing a priori attitude knowledge to Active Debris Removal missions, among others.

In most cases, the object of interest is too distant from the observer location for an optical telescope to obtain resolved images of said object – from which its characteristics could be inferred. Instead, objects appear as point- or streak-like features – these only provide the total count of photons coming from the Sun and reflected by the object. Nonetheless, when measured across several consecutive images, this same light intensity is bound to vary as a function of the object-observer-Sun relative position, as well as the object attitude, size, shape and surface reflective properties. Thus, time series of brightness measurements, a.k.a. light curves, encode these characteristics, which may be estimated by what is known as light curve inversion techniques.

This paper focuses on AD using light curves – *i.e.* non-resolved images –, under the assumption that all other fac-

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tors are known: shape, size, reflective properties, observation geometry (orbit) and telescope performance parameters. The literature on light curve inversion to determine the attitude of artificial objects in space has flourished during the past decade and a half, borrowing and adapting techniques from the asteroid characterization background, as in Hall and Kervin (2014), but also developing new ones, as in Wetterer and Jah (2009). Both batch (Piergentili et al., 2017) and sequential (see below) approaches have been explored.

This paper analyses the sequential non-linear estimation approach. Wetterer and Jah (2009) is the first work to use the Unscented Kalman filter (UKF) to estimate attitude – in Modified Rodrigues Parameters (MRP) form – and angular rates of RSOs using ground-based telescope measurements, assuming the shape and surface reflectivity properties to be known. Holzinger et al. (2012), Holzinger et al. (2014) are the first to apply the Particle filter (PF) to obtain Euler angles and attitude rates of agile objects, considering shape model uncertainty. Their approach improves upon the Extended Kalman Filter (EKF) and the UKF, which are not able to capture the high non-linearities of the problem. Coder et al. (2015) further adapt the PF approach to agile objects using a Marginalized PF, which improves the computational tractability. Additionally, they employ a Singer process noise model to account for unknown manoeuvring torques. This concept is further upgraded in Coder et al. (2017) with an improved measurement process noise model, which allows higher initial uncertainty. Coder et al. (2018) validate this implementation against the real attitude ephemerides of the Hubble Space Telescope. The PF approach is tested on uncontrolled objects to estimate Generalized Rodrigues Parameters (GRP) by Linares et al. (2014a), too. In this line, the performance of a bootstrapped PF is compared against the EKF approach on Euler angles by Bernard and Geller (2018), where they identify the extreme non-linearity of the light curve measurement model as the main reason for the EKF divergence. Du et al. (2018) improve the PF approach on GRP uncontrolled objects by using an Unscented PF, in which particles are substituted by kernels that are propagated with a bank of UKFs. Finally, Burton and Frueh (2021) introduce two alternative approaches based on the EKF and the determination of all feasible pseudo-measurements (incident-reflecting directions pairs): one pre-filters unfeasible pseudo-measurements out, to then feed the surviving combinations to the EKF; the other one uses a Probability Hypothesis Density (PHD) filter to wrap the EKF core, representing the state space as a Gaussian Mixture (GM).

The problem of sequential AD has been extended to include other states, too. Qin et al. (2018) implement a classical PF for the joint attitude-position estimation problem. Yun and Zanetti (2020) improve on this architecture by estimating the GM that best fits the probability density function (pdf) represented by the particle cloud, as per the Expectation–Maximization (EM) algorithm. They then update each kernel of the GM with the measurement using a conventional EKF/UKF core, before redrawing the next cloud of particles. Linares et al. (2014b) further extend the problem to include shape estimation. They use a bank of attitude-position UKFs embedded in a multiple-shape-model adaptive algorithm. Finally, Linares and Crassidis (2018) perform a simultaneous attitude-shape estimation by including a shape parametrization based on facet normals and albedo-area products in the state. They use a variant of the PF that proposes new high-probability samples by modelling the particles as a Hamiltonian system in probability space.

Any form of attitude estimation from light curve measurements is encumbered by the fact that both its dynamic and measurement models are significantly non-linear functions of the state, to varying degrees. Furthermore, in some scenarios the attitude state may not be fully observable, being represented by highly non-Gaussian and potentially multimodal pdfs. This is even more so if additional states are estimated simultaneously (Linares and Crassidis, 2018; Yun and Zanetti, 2020). This renders the Gaussian pdf hypothesis – typical on *e.g.* the Orbit Determination (OD) problem (Vallado and McClain, 2013) – invalid, hence the proliferation of advanced non-linear filtering techniques such as the ones summarized above. These try to 1) outperform the classical EKF/UKF, while 2) reducing the computational burden of the PF.

This paper proposes an alternative approach to the high non-linearity filtering problem of AD from light curves, in which the pdf of the state is approximated as a Gaussian Mixture, where each kernel is treated as a UKF. The key difference with similar approaches in the literature above is that it incorporates a non-linearity assessment metric, based on the Unscented Transform (UT). According to this metric, kernels are split whenever non-linearities arise, and merged back together as soon as they become too close. This approach, called the Adaptive Gaussian Mixtures Unscented Kalman Filter (AGMUKF), is enabled by the Gaussian splitting libraries developed by Vittaldev and Russell (2016). Schiemenz et al. (2020) specialized it to perform OD with air density realism. This paper adapts these two works to the AD case.

On the one hand, a fine enough GM can represent any arbitrarily complex pdf in L1 sense, something unachievable with classical EKF/UKF. On the other hand, the adaptive character of this approach strives to reduce the computational burden typical of the particle or Gaussian mixture filters – it naturally avoids the degeneration problem of the PF, as well as the need for regularization or other such techniques.

The rest of this paper is divided as follows. Section 2 outlines the non-linear sequential estimation problem, and presents the AD-specific dynamic and measurement models. Section 3 then analyses the non-linearity of these models, which motivates and justifies the Adaptive Gaussian Mixtures Unscented Kalman Filter (AGMUKF) proposed in Section 4. The AGMUKF is validated under simulated scenarios in Section 5, after which the paper closes with the final conclusions in Section 6. At the end, after the acknowledgements (Section 7) and and the literature references, several appendices provide key equations and proofs that, although not central or original to this paper, have been deemed important enough to be included and/or are not described compactly enough in any one single reference.

2. Problem definition

This section poses the problem of RSO attitude estimation using non-resolved telescope imagery, a.k.a. AD from light curves. Therefore, the conventions, hypotheses, variables, parameters and models involved are defined here. In particular, SubSection 2.1 describes the generic non-linear sequential estimation problem. It is followed by the detailed description of the attitude dynamic model (SubSection 2.2) and the measurement model (SubSection 2.3).

2.1. Non-linear sequential estimation

The generic non-linear sequential estimation problem has the goal to estimate the probability density function (pdf) of some hidden random variable $X(t) \in \mathbb{R}^n$ that varies with time and follows a continuous Markov chain, conditional to a sequence of observables $Z_k \sim \text{pdf}(z_k)$, which have been measured as z_k at times $t_k, \forall k \in [1, N_t]$ such that $k > l \Rightarrow t_k > t_l$. This is

$$X(t_{N_t}) \sim \text{pdf}(x(t_{N_t})|z_1, \ldots, z_{N_t}) . \tag{1}$$

Note that this paper assumes the convention that random variables are defined in upper case, while their realizations are written in lower case. X is often referred to as the state, while Z_k are the measurements.

Conditionality on a measurement implies conditionality on the whole sequence of measurements up to that point. Thus, from here onward,

$$\text{pdf}(\cdot|z_k) = \text{pdf}(\cdot|z_1, \ldots, z_k), \forall k \in [1, N_t] . \tag{2}$$

Furthermore, since the pdf of X is only evaluated at measurement times t_k, it is shortened as $X_k = X(t_k)$. This is, $\{X_k\}$ is a discrete Markov chain.

The initial pdf

$$X_0 \sim \text{pdf}(x_0|\varnothing) \tag{3}$$

is known, which is not conditional to any of the measurements, hence the explicit use of the empty set. The dynamic model that dictates the pdf of the state at some time t_k, conditional to the state at some other time t_l, is known, too,

$$X_{k|l} \sim \text{pdf}(x_k|x_l) , \tag{4}$$

as well as the measurement model that returns the pdf of Z_k at time t_k, given a known state x_k,

$$Z_k \sim \text{pdf}(z_k|x_k) . \tag{5}$$

The non-linear filter cycles over two steps, starting from t_1, until all measurements are processed. After the filter is initialized with

$$X_{k-1|k-1} = X_0 , \tag{6}$$

these steps are:

- **Propagation**. During the propagation step, the posterior pdf of the state from the previous cycle, $\text{pdf}(x_{k-1}|z_{k-1})$, is brought to the new measurement time t_k by marginalization, obtaining the prior

$$\text{pdf}(x_k|z_{k-1}) = \int \text{pdf}(x_k|x_{k-1})\text{pdf}(x_{k-1}|z_{k-1})dx_{k-1} . \tag{7}$$

- **Update**. In the update step, the new posterior pdf is calculated using the Bayes rule, from the new information brought by the measurement z_k:

$$\text{pdf}(x_k|z_k) = \frac{\text{pdf}(z_k|x_k)\text{pdf}(x_k|z_{k-1})}{\text{pdf}(z_k|z_{k-1})} , \tag{8}$$

where $\text{pdf}(z_k|z_{k-1}) = \int \text{pdf}(z_k|x'_k)\text{pdf}(x'_k|z_{k-1})dx'_k$ is the pdf of the measurement z_k, conditional to all previous measurements.

After having processed all the measurements, the output of the filter is the pdf of the state at the last time step, conditioned to the entire sequence of measurements (Eq. 1).

Different implementations approximate the propagation and the update steps (Eqs. 7 and 8) under different assumptions, such as Gaussianity and local linearisation (EKF), or delta-sum pdf (PF).

Apart from an initial state, any implementation of a non-linear estimation filter needs to define the dynamic and the measurement models. Such models for the AD problem are described in the following subsections.

2.2. Attitude dynamic model

The rotation operation that transforms a vector $v \in \mathbb{R}^3$ from frame A (denoted as v^A) to frame C (v^C) can be expressed as

$$\boldsymbol{v}^C = \left(\boldsymbol{q}_A^C\right)^* \boldsymbol{v}^A \boldsymbol{q}_A^C , \tag{9}$$

where $\boldsymbol{q}_A^C \in \mathbb{A}$ is the quaternion that represents such rotation; the superscript * indicates quaternion conjugate. The algebra $\mathbb{A}$ is the subspace of the quaternion space $\mathbb{H}$ that contains all the quaternions with unitary absolute value and positive real part:

$$\mathbb{A} = \{\boldsymbol{q} \in \mathbb{H} | \Re(\boldsymbol{q}) \geqslant 0 \wedge |\boldsymbol{q}| = 1\} . \tag{10}$$

Quaternions in this paper are symbolized using bold font, and when they appear in normal font they symbolize the associated 4-dimensional vector:

$$\boldsymbol{q} = [1 \quad \boldsymbol{i} \quad \boldsymbol{j} \quad \boldsymbol{k}]q . \tag{11}$$

The three hyper-complex constants are $\boldsymbol{i}$,$\boldsymbol{j}$ and $\boldsymbol{k}$, while $q \in \mathbb{R}^4$. Likewise, the bold version of a 3-dimensional vector v is its quaternion form with 0 real part. This is

$$\boldsymbol{v} = [\boldsymbol{i} \quad \boldsymbol{j} \quad \boldsymbol{k}]v. \tag{12}$$

Thus, whenever a quantity explicitly defined as a vector appears in bold, it represents the quaternion with zero real part whose three hyper-imaginary components correspond to each of the three dimensions of the vector. See Appendix A for a summary of quaternion algebra.

An attitude quaternion is related to the Euler-angle pair by

$$q_A^C = \left[\cos\left(\frac{\varphi}{2}\right) \quad e^T \sin\left(\frac{\varphi}{2}\right)\right]^T, \tag{13}$$

where $e \in \{u \in \mathbb{R}^3 | \|u\| = 1\}$ is the common rotation axis around which frame A needs to rotate φ radians to align itself with frame C. The superscript ^{T} indicates transpose, and the construct $\|\cdot\|$ is the Euclidean norm. From this, it is trivial that $\boldsymbol{q}_C^A = \left(\boldsymbol{q}_A^C\right)^*$.

In this paper, the attitude of an object is represented by the quaternion that transforms a given vector in body frame B – the frame attached to the object – to the inertial frame I – the Geocentric Celestial Reference Frame (GCRF) as defined in Kaplan (2006). This is $\boldsymbol{q}_B^I$.

With the quaternion representation of attitude, kinematics are defined by the simple equation

$$\dot{\boldsymbol{q}}_B^I = -\frac{1}{2}\boldsymbol{\omega}_{B/I}^B \boldsymbol{q}_B^I \tag{14}$$

where the dot indicates derivative over time, and $\boldsymbol{\omega}_{B/I}^B$ is the quaternion representation of $\omega_{B/I}^B \in \mathbb{R}^3$, which is the instantaneous angular velocity of frame B relative to frame I, expressed in B coordinates. See Appendix A for the derivation of this equation.

Attitude dynamics in Newtonian physics can be described by the conservation of angular momentum of a constant-mass, rigid system as

$$I_G^B \dot{\omega}_{B/I}^B = T^B - \omega_{B/I}^B \times \left(I_G^B \omega_{B/I}^B\right), \tag{15}$$

where $T \in \mathbb{R}^3$ is the net torque applied to the object, while $I_G \in \mathbb{R}^{3\times 3}$ is the tensor of inertia of the object. Without loss of generality, the rest of this paper assumes that the body frame is aligned with the principal axes of the object – *i.e.* I_G^B is diagonal. The symbol $\times$ indicates cross product.

Any non-linear filter implementation applied to attitude needs to deal with the fact that, although $q_B^I \subset \mathbb{R}^4$, because $\|q_B^I\| = 1$, there are only three degrees of freedom involved. To guarantee uniqueness of the state, the state itself should have as many dimensions as degrees of freedom. This problem is often overcome using Generalized Rodrigues Parameters (GRP), obtained from the quaternion with the map $p_{a,f} : \mathbb{A} \to \mathbb{R}^3, \forall a, f \in \mathbb{R}^+$, defined as

$$p_{a,f}(\boldsymbol{q}) = f\frac{\mathfrak{q}}{a + q_r}, \tag{16}$$

where $\mathfrak{q} = \Im(\boldsymbol{q})$ and $q_r = \Re(\boldsymbol{q})$ are the imaginary and real parts of the quaternion, respectively (see Appendix A). Whenever $f = 2(a+1)$,

$$p = e\varphi + O\left(|\varphi|^3\right). \tag{17}$$

The term $O(g(x))$ is the Landau notation that indicates that, if $g : \mathbb{R}^n \to \mathbb{R}^+$, there exist $\delta, M \in \mathbb{R}^+$ such that $|O(g(x))| \leqslant Mg(x)$ for any $0 < |x| < \delta$. Thus, this choice of f has the benefit of yielding p equal to three independent angle rotations along each axis of the frame, whenever these are small enough. See the proof in Appendix B.

The quaternion can be recovered with the inverse map $p_{a,f}^{-1} : \mathbb{R}^3 \to \mathbb{A}$, whose equations are

$$q_r = \frac{-a\|p\|^2 + f\sqrt{f^2 + (1-a^2)\|p\|^2}}{f^2 + \|p\|^2} \text{and} \mathfrak{q} = \frac{a + q_r}{f} p. \tag{18}$$

The classical UKF (Julier and Uhlmann, 2004) can then be implemented by using a reference quaternion plus an error-state based on the GRP (Crassidis and Markley, 2003; Wetterer and Jah, 2009). This is,

$$x = \begin{bmatrix} \delta p \\ \omega_{B/I}^B \end{bmatrix}, \tag{19}$$

where $\delta p = p(\delta\boldsymbol{q})$. The error-quaternion $\delta\boldsymbol{q}$ transforms from some reference body frame B_{ref} to the body frame B represented by state x:

$$\boldsymbol{q}_{B_{ref}}^B = \delta\boldsymbol{q}. \tag{20}$$

Thus, the quaternion represented by the state is

$$\boldsymbol{q}_B^I = (\delta\boldsymbol{q})^* \boldsymbol{q}_{B_{ref}}^I. \tag{21}$$

Its dynamic model is

$$x(t + \Delta t) = f\left(x(t), t, \Delta t, T^B\right) + w, \tag{22}$$

where w is white process noise (typically Gaussian) with zero mean and covariance Q, and f is the solution to the system of differential equations composed by Eqs. 14 and 15, substituting by Eqs. 16, 18 and 21 where necessary. In this work, the net torque T^B is assumed to be a known parameter of the system.

2.3. Light curve measurement model

This paper focuses on inverting light curves to estimate the attitude of an RSO. This is, the measurement model used here is the instrumental magnitude measured by an optical telescope, as defined in Vallverdú Cabrera et al. (2021).

This section presents a summary of the equations involved. The measurement used is

$$m_I = -2.5\log_{10} F_a = h\left(\boldsymbol{q}_B^I, r_i^I, r_r^I, o, s\right), \tag{23}$$

where F_a is the measured photon flux in $e^{-}s^{-1}$. This flux depends on: the sensor parameters s, which encompass the sensor transmissivity and quantum efficiency; the object shape (a closed region in space bounded by flat facets) and reflective properties o, which define how the light is reflected on its surfaces; the relative geometry between object, observer and Sun, represented by the object to Sun r_i^I and to observer r_r^I vectors – these determine the light-path solid angles, as well as the reflection angles; and finally, but most importantly for the case at hand, the attitude of the object $\boldsymbol{q}_B^I$, which determines the orientation of each of the facets of the object relative to r_i^I and r_r^I, and how the shadows propagate through non-convex geometries. For further details refer to Vallverdú Cabrera et al. (2021).

One of the cornerstones of the measurement model is the choice of the Bidirectional Reflectance Distribution Function (BRDF) that models how light reflects off the surfaces of the object. This paper uses the Cook-Torrance (CT) BRDF (Cook and Torrance, 1982), which is simple to compute, yet accurate enough for the purpose of AD, as proven by Coder et al. (2018). Additionally, it is energy-conserving, a necessary condition for a physically realistic BRDF model (Wetterer et al., 2014). The CT model extends the simple Lambertian diffuse reflection with a specular component. Its driving parameters are the diffuse ρ and specular F_0 reflectance, the proportion of diffuse reflection d, and a slope parameter m that defines the shape of the specular component along the hemisphere of possible incident/reflected directions ($0 \leqslant \rho, F_0, d \leq 1$ and $m > 0$). Refer to Wetterer et al. (2014) for a detailed development of the Cook-Torrance BRDF. The CT equations, like any other physically realistic BRDF, are non-linear with respect to the incident/reflected directions of each facet.

Another driving factor of the measurement model is the shape of the object and the effect of self-shadowing, which can be costly to compute for non-convex objects. This paper uses the shadow projection algorithm developed in Vallverdú Cabrera et al. (2021) to solve this problem. For small changes in r_i^B and r_r^B, sharp edges and other shape features can easily result in abrupt changes in the reflected intensity.

Therefore both the BRDF model and the sensitivity to complex shapes, as well as the logarithm in Eq. 23, make the function $h\left(\boldsymbol{q}_B^I, \dots\right)$ highly non-linear against the attitude of the object.

To conclude, the measurement space for non-linear filtering is

$$z = [\, m_I \,], \tag{24}$$

with m_I being the instrumental magnitude of the object measured at the telescope. The hypothesis taken in this paper is that the overall noise effects in the measurement can be modelled as Gaussian noise. Thus, the measurement estimation model is ruled by

$$z(x) = h\left(\boldsymbol{q}\left(\delta p, \boldsymbol{q}_{B_{ref}}^I\right), \dots\right) + v\,. \tag{25}$$

The function $\boldsymbol{q}\left(\delta p, \boldsymbol{q}_{B_{ref}}^I\right)$ symbolizes the process of obtaining the attitude quaternion $\boldsymbol{q}_B^I$ represented by δp (the first three components of x), by means of Eqs. 18 and 21. To simplify the notation from here onward, this paper shall use $h\left(\delta p, \boldsymbol{q}_{B_{ref}}^I\right) \equiv h\left(\boldsymbol{q}\left(\delta p, \boldsymbol{q}_{B_{ref}}^I\right), \dots\right)$. The sensor model noise is represented by $V \sim \mathcal{N}(v|0, R)$, where $R_k := R$ is equal for all measurements. Finally, the dots represent the rest of parameters from Eq. 23, assumed to be known.

3. Motivation

The purpose of this section is to analyse the non-linearities introduced by the dynamic and measurement models of the AD problem defined above, and to pave the way for an efficient non-linear filter implementation that is aware of the degree of non-linearity at all times.

For the sake of clarity, the following shorter notations shall be used form here onward: $\boldsymbol{q} \equiv \boldsymbol{q}_B^I, \tilde{\boldsymbol{q}} \equiv \boldsymbol{q}_{B_{ref}}^I$ and $\omega \equiv \omega_{B/I}^B$.

3.1. The non-linearity index

Vittaldev and Russell (2016) propose a non-linearity index (NLI) to assess the the degree in which an arbitrary transformation deviates from linearity along a given direction, by means of a numerical approximation of the second order derivative of the transformed space against the original one. Schiemenz et al. (2020) adapt the concept to exploit the sigma points of an Unscented Transform (UT) – see Appendix C for a detailed description of the UT.

Let us define a random variable in $X \in \mathbb{R}^n$ that follows a multivariate normal distribution, $X \sim \mathcal{N}(x|\mu_x, P_x)$, whose mean vector and covariance matrix are encoded in a sigma cloud $\mathcal{X}$, as defined by Eq. C.2. Let $g : \mathbb{R}^n \to \mathbb{R}^m$ be a map that applies a non-linear transformation to x – *e.g.* let it model a physical process whose associated uncertainty is $V \sim \mathcal{N}(v|0, P_v)$. Being $\mathcal{Y}$ the sigma cloud transformed under g,

$$\mathcal{Y}(i) = g(\mathcal{X}(i)), \forall i \in [0, 2n]\,, \tag{26}$$

which represents the pdf of another random variable Y (under UT assumptions), one can asses the non-linearity introduced by g along each independent direction of the original sigma cloud with the non-linearity index $\phi \in \mathbb{R}^n$.

$$\phi_i = \left\| \frac{\mathcal{Y}(i) + \mathcal{Y}(i+n) - \mathcal{Y}(0)}{2(n+\lambda)} \right\|, \forall i \in [1, n]\,, \tag{27}$$

where λ is the parameter of the UT that controls the size of the sigma cloud (see Eq. C.3). Each dimension i of ϕ, ϕ_i, corresponds to the non-linearity index associated to the direction

$$u_i = \frac{S_i}{\|S_i\|}\,, \tag{28}$$

in the original x space, where S_i is the i-th column of the Cholesky decomposition of $P_x = SS^T$. This is, each direction u_i is the direction along which the pair of sigma points $\mathcal{X}(i)$ and $\mathcal{X}(i+n)$ have been spanned, on opposite sides of $\mathcal{X}(0)$.

One problem of this approach is that, when the characteristic dimensions of the state differ between independent directions, it is difficult to interpret ϕ unless there is some canonical rescaling that can be applied to $\mathcal{Y}$. The work of Schiemenz et al. (2020) uses a scaling *ad hoc* to the OD problem. In the present work, however the NLI is further modified to circumvent this problem with an elegant and general approach by applying Eq. 27 to the standardized sigma cloud

$$\bar{\mathcal{Y}}(i) = T^{-1}\mathcal{Y}(i), \forall i \in [0, 2n], \tag{29}$$

where $P_y = TT^T$ is the covariance of Y as estimated by the UT, and T its Cholesky decomposition. This is, if Y were normally distributed as $Y \sim \mathcal{N}(y|\mu_y, P_y)$, then $\overline{Y} \sim \mathcal{N}(\bar{y}|T^{-1}\mu_y, I_{m\times m})$.

Using Eq. 29, one rescales the transformed sigma cloud so that each dimension of $\overline{Y}$ is associated to each principal direction of P_y, and each of these directions is given the same importance within the NLI. This re-interpretation of the NLI still assesses non-linearity by to what extent lines with constant metric in x space become curves with varying metric in y space, but it considers two independent directions equally non-linear if their transformed curves have the same shape relative to the uncertainty associated to each of them. In practice, this could mean that even for transformations where P_x is very small, high NLI values could rise if g is strongly non-linear. However, the fact that the transformed P_y includes the model noise P_v, even when P_x is small, any curvatures in y of a scale smaller than the error volume of P_v will yield a relatively low NLI.

With this new NLI, one can compare the non-linearity, not only between two directions with different characteristic sizes for one given transformation, but also between different models applied to the same random variable. In the case at hand, for instance, it is possible to compare how much non-linearity is introduced by the dynamic (f) and the measurement (h) models relative to each other.

Simple, yet non-linear test functions g (*e.g.* high order polynomials, exponentials, sines and logarithms) have been explored. Visually, the density histogram of a transformed Monte Carlo sample starts to differ from Gaussianity for values of $\phi \sim 0.1$, while maximum observed values were around $\phi = 1$. These thresholds, however, still depend on the particularities of each transformation g, and possibly on the size of its input and output spaces.

3.2. Non-linearity analysis of the attitude determination problem

This subsection focuses on the analysis of the non-linearities in the problem of AD, using the Non-Linearity Index (NLI or ϕ) defined in the previous subsection.

Within the AD problem exposed in the previous section, simulation results suggest that the highest non-linearity contribution to the filtering cycle arises through the measurement model, not through the dynamic model. This is to be expected, given the highly non-linear characteristics of the measurement model (Bernard and Geller, 2018).

Fig. 1 shows the evaluation of the NLI along one UKF run on the scenario used by Wetterer and Jah (2009), the details of which are presented below in Section 5.1. Each point is the NLI in the maximum-NLI direction – *i.e.* $\max(\{\phi_i\})$. While blue + marks are the non-linearity computed for the dynamic model (propagation step), orange $\times$ marks are those that result from the measurement model (update step). It is clear from the plot that the measurement update has a maximum NLI of at least two orders of magnitude higher than the prediction step, for this case.

Focusing on the measurement update step, Fig. 2 shows the estimation of the probability distribution of the measurement for the cases of $k = 1, \ldots, 5$ of the Wetterer and Jah (2009) scenario. In these plots, the gray histogram is a Monte Carlo (MC) approximation of the pdf of the estimated measurement obtained by drawing 100 random samples from $\mathcal{N}(x_k|\hat{x}_{k|k-1}, P_{k|k-1})$, where $\hat{x}_{k|k-1}$ and $P_{k|k-1}$ are the prior mean vector and covariance matrix of the state – the histogram contains the magnitude estimated from each random sample. Then, the dashed green curve is the pdf $\mathcal{N}(z_k|\tilde{z}_k, R_k)$, a Gaussian centred around the measured (sample) $\tilde{z}_k$ with uncertainty covariance R_k. Finally, the solid orange one is $\mathcal{N}(z_k|\hat{z}_k, P_k^{zz} - R_k)$, where $\hat{z}_k$ and P_k^{zz}, are the mean and covariance obtained from applying the UT with the measurement model to $\hat{x}_{k|k-1}$ and $P_{k|k-1}$. In these figures, it is obvious that the UT-estimated pdf (orange solid curve) and the more realistic MC-estimated one (gray histogram) are in significant disagreement.

To sum up, the goal of this paper is to define a non-linear adaptive GM filter architecture that focuses on dealing with the measurement model non-linearity.

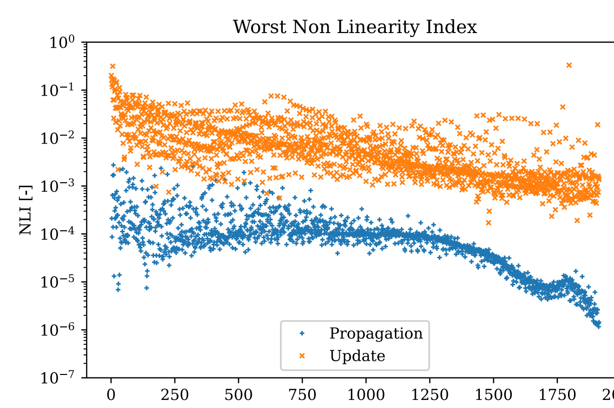


Fig. 1. Non-Linearity Index of one UKF run, based on the scenario from Section 5.1.

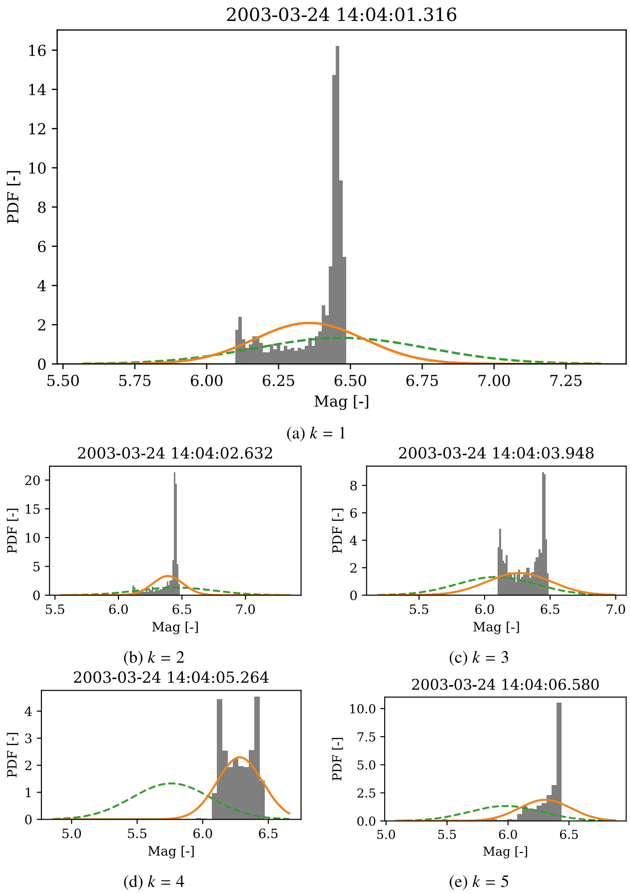


Fig. 2. Probability distribution function of $\hat{z}_k$ for the first five measurements in the Wetterer and Jah (2009) scenario when using the UKF. Legend: pdf of the observation (dashed green line); pdf of the measurement estimated from the state using UT (solid orange line) or Monte Carlo (gray histogram). Dates in the titles expressed in ISO format: YYYY-MM-DD hh:mm.ss.fff (f stands for millisecond).

4. Methodology

This section defines the Adaptive Gaussian Mixtures Unscented Kalman Filter (AGMUKF) for Attitude Determination (AD). The algorithm proposed below adapts the idea introduced by Vittaldev and Russell (2016), which Schiemenz et al. (2020) apply to the classical Orbit Determination (OD) use case with density uncertainty realism. The AGMUKF version developed in the present paper focuses on capturing the high non-linearity introduced by the measurement model, as justified by the non-linearity analysis above.

The working principle of the AGMUKF is to estimate the state pdf as a mixture of Gaussian distributions, and then to apply normal UKF steps to each member of the mixture. In contrast to the classical EKF/UKF, which represent the state pdf as a single Gaussian, the mixture approach can represent arbitrarily complex pdfs. Furthermore, the AGMUKF is labelled as *adaptive* because it exploits 1) the NLI defined in Eq. 27 to refine the mixture, together with 2) a distance metric to coarsen it. Its aim is for the mixture to always approximate the state pdf as faithfully as possible, yet with the minimum number of underlying Gaussian kernels. Thus, careful analysis of the problem is necessary to decide where in the filter steps refinement and coarsening should happen. If this is done right, the AGMUKF may be more efficient than other complex-pdf-modelling filters, such as the PF and its variants (Du et al., 2018; Coder et al., 2017).

In the OD case of Schiemenz et al. (2020), the major non-linearity contribution happens within the dynamic model, where orbit propagation over long spans of time is coupled with realistic modelling of density uncertainty. This motivates the refinement and coarsening steps to alternate with consecutive propagation steps (without updates in-between). In this case, the yardstick used to decide whether and how to split is the NLI of dynamic model.

In the AD problem, however, non-linearity comes mainly form the measurement model, and propagation time spans are relatively short for a dense light curve. Therefore, the implementation of the AGMUKF for AD introduced here refines the mixture according to the measurement model NLI, instead, and does so just before the measurement update step. This way, the measurement pdf is modelled with finer detail. Right after the update, the filter merges the mixture back, so that the state pdf does not remain oversampled.

Following, each AGMUKF step is described in detail. Of these, The UKF steps of the AGMUKF are mainly adapted from and inspired by Crassidis and Markley (2003, 2009, 2018).

4.1. Probability distribution

Following the non-linear filtering notions introduced in SubSection 2.1, the AGMUKF estimates the pdf of the state X as a Gaussian Mixture (GM) with M kernels:

$$X \sim \mathrm{pdf}(x) = \sum_{j=1}^{M} w^j \mathcal{N}(x|\mu^j, P^j)\,, \tag{30}$$

where w^j is the weight of kernel j, which follows a normal distribution with mean vector μ^j and covariance matrix P^j. Note that $w^j = \mathrm{pdf}(\eta = j)$ is the value of the probability mass function (pmf) of a discrete random variable $H \sim p(\eta)$, with $\eta \in [1, M]$. Thus,

$$\sum_{j=1}^{M} w_j = 1\,. \tag{31}$$

Furthermore, $\mathrm{pdf}(x|\eta = j) = \mathcal{N}(x|\mu^j, P^j)$, so the expression in Eq. 30 is just $\mathrm{pdf}(x)$ obtained through marginalization.

The first and second statistical moments of a GM are, then

$$\mu = \sum_{j=1}^{M} w^j \mu^j \tag{32}$$

and

$$P = \sum_{j=1}^{M} w^j \left[P^j + (\mu^j - \mu)(\mu^j - \mu)^T \right] \tag{33}$$

respectively (Schiemenz et al., 2020).

4.2. Filter initialization

The filter is initialized with the known pdf of the state at some time t_0, represented by the reference quaternion $\tilde{\boldsymbol{q}}_0$, and initial GM with M_0 kernels represented by

$$X_0 \sim \sum_{j=1}^{M_0} w_0^j \mathcal{N}\left(x_0 | \hat{x}_0^j, P_0^j\right), \tag{34}$$

where $\hat{x}_0^j$ and P_0^j are the a priori estimated means and covariances of the state. Each kernel of the filter is initialized as in the conventional Unscented Kalman Filter (UKF) algorithm. This is, the initial sigma cloud of each kernel, $\mathcal{X}_0^j$, composed by $2n+1$ points in state space (where $n = 6$) is obtained using Eq. C.2.

The rest of the steps are executed in a loop until all measurements have been processed.

4.3. Propagation step

For each new measurement sample $\tilde{z}_k$, measured at t_k, the state pdf first needs to be propagated from the previous measurement update, at t_{k-1}, to t_k. This is, this subsection deals with obtaining the prior pdf $\mathrm{pdf}(x_k|z_{k-1})$ from the last known posterior $\mathrm{pdf}(x_{k-1}|z_{k-1})$.

To do so, first, the sigma cloud of each kernel is propagated using the dynamic model defined by the map $f : \mathbb{R}^n \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}^n$ (Eq. 22). To do so, the δp component of each sigma point needs to be converted back to quaternion state using Eqs. 18 and 21, obtaining $\boldsymbol{q}^j_{k-1|k-1}(i)$. With this, each sigma point is propagated from t_{k-1} to t_k using Eqs. 14 and 15 and a numerical integration routine (Shampine and Gordon, 1975), which yields $\boldsymbol{q}^j_{k|k-1}(i)$.

The prior reference quaternion needs to be redefined, so that the sigma cloud of each kernel, whose points are in quaternion form, can be converted back to state space – *i.e.* the GRP form. To do so, one propagates the first statistical moment of the previous step, a.k.a. the merged mean $\hat{x}_{k-1|k-1}$ (obtained from the set $\left\{w^j, \hat{x}^j_{k-1|k-1}\right\}$ with Eq. 32) to t_k, using the same method of GRP-to-quaternion as with the sigma points. This yields the merged, propagated attitude quaternion estimate $\boldsymbol{q}_{k|k-1}$, which becomes the new reference quaternion:

$$\tilde{\boldsymbol{q}}_{k|k-1} = \boldsymbol{q}_{k|k-1}. \tag{35}$$

With the new reference, the sigma cloud $\mathcal{X}_k^j$ of each kernel is recovered by using the inverse of Eq. 21 followed by Eq. 16 to obtain the δp representation of each $\boldsymbol{q}^j_{k|k-1}(i)$.

At this point, the estimated prior mean $\hat{x}^j_{k|k-1}$ and covariance $P^j_{k|k-1}$ of each kernel are obtained with the conventional UT: Eqs. C.5 and C.6 are used on the propagated cloud $\mathcal{X}_k^j$, using the process noise covariance matrix of step k, Q_k, as model noise. Note that for the clouds to include the additional process noise covariance Q_k, they need to be redrawn using Eq. C.2 (Wu et al., 2005). Under the assumption that all kernels are affected by the same process noise, Q_k can be added to each one of them with the overall effect of increasing the merged covariance by the same amount (Schiemenz et al., 2020).

Finally, assuming the propagation is linear enough, the kernel weights can remain constant (Schiemenz et al., 2020):

$$w^j_{k|k-1} = w^j_{k-1|k-1}. \tag{36}$$

Likewise, the number of kernels does not change either, so $M_{k|k-1} = M_{k-1|k-1}$. With this, the propagated prior state pdf is

$$X_{k|k-1} \sim \sum_{j=1}^{M_{k|k-1}} w^j_{k|k-1} \mathcal{N}\left(x_k | \hat{x}^j_{k|k-1}, P^j_{k|k-1}\right). \tag{37}$$

4.4. Measurement estimation

The traditional update step of the UKF, where the measurement pdf is used to update the prior state pdf $\mathrm{pdf}(x_k|z_{k-1})$ to the posterior $\mathrm{pdf}(x_k|z_k)$, has been split into multiple steps, to accommodate the Gaussian splitting and merging strategies that characterize the AGMUKF for AD.

This section describes the first part of the UKF update step, which consists in applying the measurement model on the sigma cloud of each kernel, using the map $h : \mathbb{R}^3 \times \mathbb{A} \to \mathbb{R}$ from Eq. 25. This yields the measurement sigma clouds $\mathcal{Z}_k^j$. Next, the means $\hat{z}_k^j$ and innovation covariance matrices $P_k^{zz,j}$ are calculated using again the UT (Eqs. C.5 and C.6), this time applied on $\mathcal{Z}_k^j$ and using the measurement noise covariance matrix R_k to represent model noise.

This step finishes with an estimation of the innovation pdf,

$$\mathrm{pdf}(z_k|x_k) = \sum_{j=1}^{M_{k|k-1}} w^j_{k|k-1} \mathcal{N}\left(z_k | \hat{z}_k^j, P_k^{zz,j}\right). \tag{38}$$

4.5. Refinement step

Because of the high non-linearity of h, before updating the state with the measurement information, it is necessary to check whether the innovation pdf has become too non linear, in which case the prior pdf should be split into a finer GM, to minimize the rise of non-linearities in the measurement estimation step from SubSection 4.4.

To do so, the NLIs of the measurement model in each kernel are computed from each sigma cloud $\mathcal{Z}_k^j$ with Eq.

27, using $P_k^{zz,j}$ as the scale term in Eq. 29. These NLIs are used to decide which kernels should be split, into how many parts, and along which directions.

The split library used is the one developed by Vittaldev and Russell (2016), which is homoscedastic – meaning that the resulting new kernels have the same common covariance. This library allows splitting one kernel among up to 39 new ones, only in odd numbers.

Following the implementation in Schiemenz et al. (2020). The split recommendation for kernel j in each direction i in state space takes the form

$$n_i^j = \text{odd}\left(a\left(\log_{10}\left(\phi_i^j\right) + b\right)\right), \tag{39}$$

where $\text{odd}(x) = 2\lceil x/2 \rceil - 1$.

The splitting library works in the standard normal univariate distribution. For a split of L new kernels, it provides the set of weights and means $\left\{\bar{w}_{l|L}, \bar{m}u_{l|L} | l \in [1, L]\right\}$, as well as the covariance $\bar{\sigma}_L^2$, that result in the symmetric univariate GM with merged mean 0 and covariance 1 – *i.e.* the moments of the standard normal distribution. Therefore, to split along the direction $u \equiv u_i^j$ (Eq. 28) from kernel j with weight $w = w_{k|k-1}^j$, mean $\mu \equiv \hat{x}_{k|k-1}^j$ and covariance $P \equiv P_{k|k-1}^j$, first the original multivariate normal of the kernel needs to be projected onto the standard normal along direction u. This is done by linearly transforming the state using *e.g.* the Cholesky decomposition S of $P = SS^T$ (Vittaldev and Russell, 2016), so that

$$\bar{u} = \frac{S^{-1}u}{\left\|S^{-1}u\right\|}. \tag{40}$$

Then, the $L \equiv n_i^j$ new kernels are obtained as the sequence of new weights

$$w_l = w \cdot \bar{w}_{l|L}, \tag{41}$$

means

$$\mu_l = \mu + \bar{\mu}_{l|L} S \bar{u} \tag{42}$$

and covariances

$$P_l = S\left(I + \left(\bar{\sigma}_L^2 - 1\right)\bar{u}\bar{u}^T\right)S^T \tag{43}$$

for all $l \in [1, L]$, where I is the $n \times n$ identity matrix.

After the split, all the newly created kernels need to have their sigma cloud $\mathcal{X}_k^j$ redrawn (Eq. C.2), so that the measurement estimation step from SubSection 4.4 can be applied on them, too.

If several directions in the same kernel are recommended for refinement, the split operation can be applied first in one direction, and then in the other one to the new kernels. As long as the directions are orthogonal, the order in which these are applied is irrelevant (Schiemenz et al., 2020).

Now, the prior and innovation pdf equations are equal to Eq. 37 and 38, with the difference that there are $M'_{k|k-1} \geqslant M_{k|k-1}$ kernels. This results in a denser mixture that captures the pdf in measurement space with significantly higher accuracy, even if the pdf in state space remains virtually the same as before the split.

4.6. Update step

Having split the state pdf to mitigate the appearance of non-linearities in each UKF kernel, one can proceed with the normal UKF steps to obtain the posterior pdf of the state, $\text{pdf}(x_k|z_k)$.

On a kernel basis, the UKF cross covariance $P_k^{xz,j}$ is computed by applying Eq. C.8 on the state $\mathcal{X}_k^j$ and measurement $\mathcal{Z}_k^j$ sigma clouds, in this order. With this, one obtains the Kalman gains

$$K_k^j = P_k^{xz,j}\left(P_k^{zz,j}\right)^{-1} \tag{44}$$

and updates the kernel means and covariances with

$$\hat{x}_{k|k}^j = \hat{x}_{k|k-1}^j + K_k^j\left(\tilde{z}_k - \hat{z}_k^j\right) \tag{45}$$

and

$$P_{k|k}^j = P_{k|k-1}^j - K_k^j P_k^{zz,j}\left(K_k{}^j\right)^T. \tag{46}$$

In the update step, actual new information has been introduced in the state pdf. Thus, the kernel weights must be updated. This is done by rescaling each kernel weight with the innovation likelihood of the prior residual (Schiemenz et al., 2020),

$$w_{k|k}^j = w_{k|k-1}^j \mathcal{N}\left(\tilde{z}_k | \hat{z}_k^j, P_k^{zz,j}\right). \tag{47}$$

Now, the posterior pdf looks like

$$X_{k|k} \sim \sum_{j=1}^{M'_{k|k-1}} w_{k|k}^j \mathcal{N}\left(x_k | \hat{x}_{k|k}^j, P_{k|k}^j\right). \tag{48}$$

4.7. Coarsening step

After the measurement update, it is often the case that the Gaussian mixture that represents the posterior state pdf is oversampled, due to all the extra kernels added to the mixture to properly model the non-linearity in the measurement estimation. Thus, a merge step is performed, where kernels that are too close to each other in state space are joined within a moment-preserving merge.

The moment-preserving merge of any two kernels i and j in the mixture is the weight-normalized version of Eqs. 32 and 33 applied to the two kernels only – the subscript $_{k—k}$ of weights, means and covariances has been omitted for brevity since it applies everywhere in this subsection:

$$w^{ij} = w^i + w^j, \tag{49}$$

$$\mu^{ij} = \frac{w^i \mu^i + w^j \mu^j}{w^{ij}} \tag{50}$$

and

$$P^{ij} = \frac{w^i P^i + w^j P^j}{w^{ij}} + \frac{w^i w^j}{\left(w^{ij}\right)^2}\left(\mu^i - \mu^j\right)\left(\mu^i - \mu^j\right)^T. \tag{51}$$

To coarsen the mixture, the same strategy used by Schiemenz et al. (2020) is used here. This is, the first step is to compute the Kullback–Leibler (KL) distance (Runnalls, 2007) between all kernel pairs (i,j),

$$B(i,j) = \frac{1}{2}\left[w^{ij}\log(|P^{ij}|) - w^i\log(|P^i|) - w^j\log(|P^j|)\right]. \quad (52)$$

Then, if the closest pair is at a distance lower than a given threshold, the two affected kernels are substituted by the new merged one. The distance of the new kernel relative to all the other kernels is computed anew, and the cycle is repeated until the closest pair is at a distance greater than the threshold. Refer to Schiemenz et al. (2020) for the reasons to choose the KL from all the distance metrics available in the literature.

At the end of the merge step, the posterior pdf looks similar to Eq. 48, but with $M_{k|k} \leqslant M'_{k|k-1}$ kernels due to the merge.

4.8. Filter reset

To close the filtering cycle, the posterior reference quaternion needs to be updated again. To do so, the merged mean of the posterior state pdf, $\hat{x}_{k|k}$, is computed as in Eq. 32, from which the posterior merged error GRP, $\delta p_{k|k}$, is extracted. The merged error quaternion $\delta\boldsymbol{q}_{k|k}$ is obtained using Eq. 18 on $\delta p_{k|k}$. Similar to the propagation step, the posterior reference is computed from the prior one as

$$\tilde{\boldsymbol{q}}_{k|k} = \delta\hat{\boldsymbol{q}}^*_{k|k}\tilde{\boldsymbol{q}}_{k|k-1}. \quad (53)$$

With the new reference, the kernel means need to be shifted, too. In UKF implementations for attitude states with reference quaternion plus error state (Crassidis and Markley, 2003), this is commonly achieved by resetting them with the merged error GRP:

$$\delta\hat{p}^j_{k|k} \leftarrow \delta\hat{p}^j_{k|k} - \delta p_{k|k}. \quad (54)$$

This process makes the merged mean of the mixture yield exactly $\delta p_{k|k} = 0$, meaning that the reference quaternion is exactly the one represented by the merged mean of the state. Eq. 54, however, is only a first order approximation (see Appendix B) that is only valid if both $\delta\hat{p}^j_{k|k}$ and $\delta p_{k|k}$ are very small. This is appropriate for filters like the plain UKF, which assume small uncertainties relative to the total surface of the unit quaternion hypersphere. However, the pdf of the state within the AGMUKF may be composed, at any given step, by a disperse mixture spanning across wide regions of the attitude space. This means that the errors introduced by Eq. 54 are likely to be significant if used within the AGMUKF, especially when the second statistical moment of the mixture is relatively high.

The proposed solution is to exploit the UKF kernel of the AGMUKF to reset the state of each kernel relative to the new reference quaternion by means of an UT, so that at least the second order statistical moment of each kernel is preserved. To do so, first, the sigma cloud of each kernel is redrawn using Eq. C.2 from its mean $\hat{x}^j_{k|k}$ and and covariance $\widehat{P}^j_{k|k}$. Then, each sigma point i of each kernel j is modified as

$$\delta p^j_{k|k}(i) \leftarrow p\left(\left(p^{-1}\left(\delta p_{k|k}\right)\right)^* p^{-1}\left(\delta p^j_{k|k}(i)\right)\right), \quad (55)$$

which is the actual quaternion composition that ensures that each sigma point still represents the same quaternion when composing it with the new reference quaternion. Then, the mean and covariance of each kernel are recovered using the normal UT from Eqs. C.5 and C.6, as was done during the propagation step. Due to the nonlinearity of this operation, the new merged mean will not yield exactly $\delta p_{k|k} = 0$; but because it will still approach Eq. 54 for small error states, it will in general imply a reduction of $\|\delta p_{k|k}\|$ proportional to second order error – *i.e.* the reference quaternion will still be a closer representation to the updated merged state than before the reset.

5. Simulation results

In this section, different scenarios of the AD problem are presented and discussed. It is worth mentioning that all the results shown in this section have been obtained using the square-root version of the Unscented Kalman Filter (SR-UKF), which is mathematically equivalent to the plain UKF steps presented in the previous section, but guarantees the positive-definiteness of the covariance matrix. It synergises well with the AGMUKF, since the Cholesky decompositions of the covariance matrices are needed anyway to establish the split directions. Refer to Van der Merwe and Wan (2001) for a detailed development of the core SR-UKF.

Furthermore, all cases presented here use $a = 1$ and $f = 2(a+1) = 4$ for the GRP implementation, as *e.g.* in Du et al. (2018).

5.1. AGMUKF vs. UKF

The first validation step of the AGMUKF has been to compare it against a plain UKF, which is in this case equivalent to the AGMUKF algorithm presented above, with only one kernel and skipping the splitting and merging steps, plus using Eq. 54 to reset the error state.

The simulated scenario reproduces the exact same case presented in Wetterer and Jah (2009). This is, the Atlas-Centaur II upper-stage with NORAD number 12069 is simulated based on its Two-Line Element set (TLE), available from Space-Track.org (U.S. Government, 2019). The initial attitude, angular velocity, measurement acquisition rate and simulation time span are unchanged w.r.t. Wetterer and Jah (2009). The shape model conserves the same characteristics: a cylinder with 9m length, 1.5m radius, and two caps on either side. In the current work, the caps are modeled as cones of 1.5m height, instead of flattened hemispherical ones (see Fig. 3). Both cylinder and cones are discretized using 100 flat facets each. As in Wetterer and Jah (2009), the sides of the cylinder (blue) follow a Cook Torrance (Wetterer et al., 2014) BRDF with $\rho = F_0 = 0.2$ and $d = 0.3$, with $m = 0.17$. The cones, also Cook Torrance, have the same m and $d = 0.8$; one cone has $\rho = F_0 = 0.6$ (orange), while the other one has $\rho = F_0 = 0.3$ (green). The inertia tensor is computed assuming the object is solid, with constant density. Regarding the filter, the same measurement model uncertainty, initial state uncertainty and process noise are used. Finally, the simulated observer is the AEOS telescope.

We have tried to reproduce the Wetterer and Jah (2009) scenario as faithfully as possible, not only to assess the AGMUKF against a known-working case of the UKF, but so that our own UKF implementation can be validated, since it is the work-horse underneath the AGMUKF.

The UKF (as well as the AGMUKF) has been tuned with $\alpha = 0.5, \beta = 2$ and $\kappa = 0$. Regarding the parameters only relevant to the AGMUKF, the split criteria from Eq. 39 have been set to $a = 15$ and $b = 1$. The KL-distance threshold has been set to 10^{-4} – *i.e.* kernel pairs at a relative distance of less than that value are merged in the merge step. These values have been obtained empirically by trying to balance the number of kernels and filter performance. Additionally, the maximum number of kernels allowed after any refinement step is set to $M_{\max} = 999$. Whenever a split step would generate more than that, the following measures are taken:

1. The kernels are split by order of decreasing weight. This ensures that the splitting effort is focused on the kernels with the highest likelihoods.
2. While splitting any particular kernel j, if it would exceed the maximum number of total kernels, the split recommendation vector is reduced by 2 (*i.e.* $n_i^j \leftarrow \max\left(n_i^j - 2, 0\right)$) iteratively until the split operation does not generate more kernels than the threshold, or until all $n_i^j = 0$, in which case the kernel is skipped from splitting. The jumps of 2 by 2 arise from the fact that the splitting libraries of Vittaldev and Russell (2016) only allow odd-numbered splits.

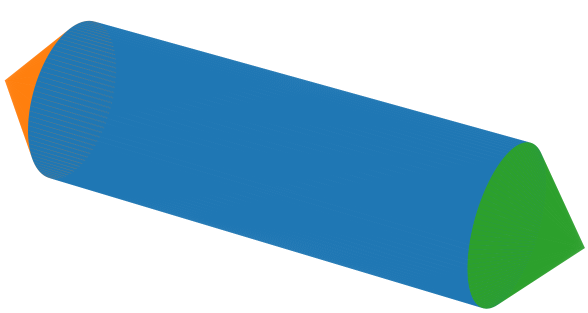


Fig. 3. Simulated object shape.

3. The kernel splitting step is stopped as soon as the mixture has $M_{\max} - 1$ kernels, since the minimum possible split would add 2 new kernels as per the libraries of Vittaldev and Russell (2016).

From the three available splitting libraries (Vittaldev and Russell, 2016), the one that yields the smallest individual kernel covariance is used ($\sigma^2 = M^{-1}$), because it is the one that can split a kernel in more pieces (39), and because the purpose of splitting is to reduce the uncertainty region of each kernel, so that the subsequent Unscented Transforms are closer to local linearity.

Having set the scenario, a Monte Carlo study of 100 iterations is performed. In each Monte Carlo iteration, both a UKF and an AGMUKF are initialized with the same random realization $\hat{x}_0$ from $\widehat{X}_0 \sim \mathcal{N}(\hat{x}_0|\mu_0, P_0)$, where P_0 is the one used in Wetterer and Jah (2009),

$$P_0 = \mathrm{diag}\left(\left[4\cdot10^{-2}\quad 4\cdot10^{-2}\quad 4\cdot10^{-2}\quad 10^{-24}\quad 10^{-24}\quad 10^{-6}\right]\right), \tag{56}$$

while

$$\mu_0 = \left[0\quad 0\quad 0\quad 0\quad 0\quad 0.252\,\tfrac{\mathrm{rad}}{\mathrm{s}}\right]^T. \tag{57}$$

The quaternion represented by the $\delta\hat{p}_0$ portion of $\hat{x}_0$ is set as the initial reference quaternion,

$$\tilde{\boldsymbol{q}}_0 = (q(\delta\hat{p}_0))^* \boldsymbol{q}_0\,, \tag{58}$$

after which $\delta\hat{p}_0$ is reset to 0. The quaternion

$$\boldsymbol{q}_0 = 0.743 - 0.171\boldsymbol{i} + 0.050\boldsymbol{j} - 0.645\boldsymbol{k} \tag{59}$$

is the true initial attitude quaternion of the object. The Wetterer and Jah (2009) covariance P_0 is used to initialize the filter – the AGMUKF starts with a single kernel. Measurements $\tilde{z}_k$ are likewise realizations of $Z_k \sim \mathcal{N}(z_k|z(t_k), R_k)$, where $z(t_k)$ is the true measurement in the absence of sensor noise at time t_k, obtained by propagating the initial true attitude and angular velocities to t_k with Eqs. 14 and 15, and then applying the measurement model from Eq. 25 with $v = 0$. Likewise, the filter uses R_k as the measurement covariance, too.

The first part of this subsection focuses on the analysis of one arbitrary filter run whose initial reference quaternion and angular velocity are

$$\tilde{\boldsymbol{q}}_0 = 0.756 - 0.070\boldsymbol{i} + 0.237\boldsymbol{j} - 0.605\boldsymbol{k} \tag{60}$$

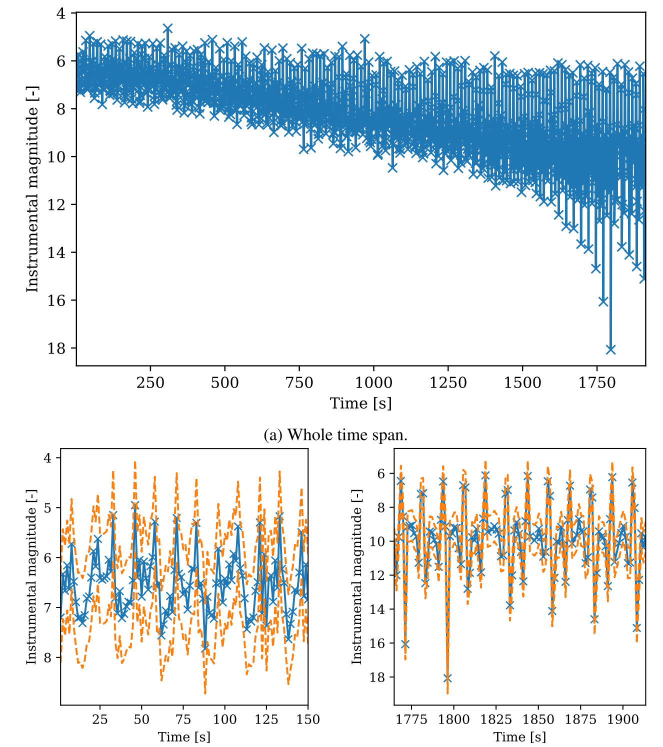
and

$$\hat{\omega}_0 = \left[-1.10\cdot10^{-12}\quad -7.79\cdot10^{-13}\quad 0.253\right]^T \frac{\mathrm{rad}}{\mathrm{s}}. \tag{61}$$

Both simulation and filter assume zero net torque, and the filter assumes a process noise of (Wetterer and Jah, 2009)

$$Q_k = \mathrm{diag}\left(\left[4\cdot10^{-8}\quad 4\cdot10^{-8}\quad 4\cdot10^{-8}\quad 10^{-24}\quad 10^{-24}\quad 10^{-12}\right]\right). \tag{62}$$

The measured (*i.e.* noisy) light curve is shown in Fig. 4. In the three plots, the solid blue line connects the measure-



(a) Whole time span.

(b) First 3 minutes.

(c) Last 3 minutes.

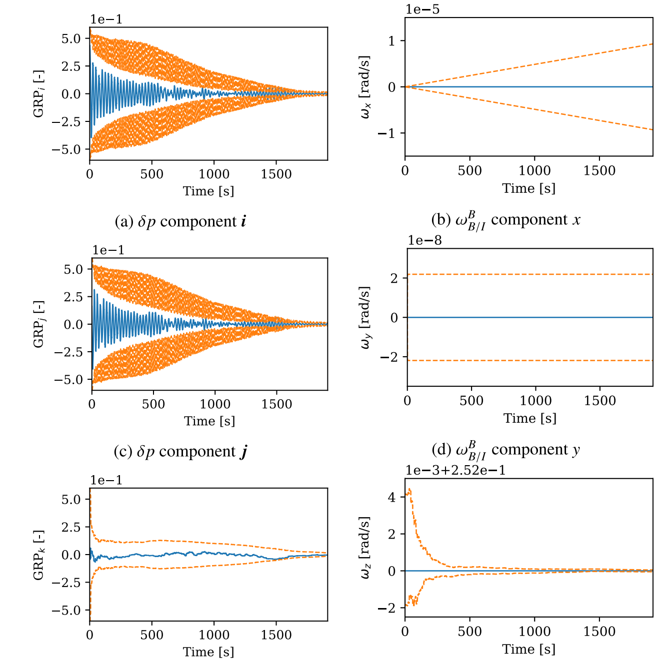
Fig. 4. One noisy light curve of the Wetterer and Jah (2009) scenario. Blue crosses are brightness measurements, while the blue line is a visual guide. The orange dashed line is the 3-σ envelope of uncertainty around the light curve.

ments, which are marked with a $\times$ symbol. While Fig. 4a shows the whole simulation time span, Figs. 4b and 4c contain the first and last 3 min, respectively, so that finer detail can be appreciated. The dashed orange curves symbolize the upper and lower bounds of the regions under which the the cumulative probability of the measurement is about 99.73%. For the measurement, which follows a Gaussian distribution with covariance $R = 0.3^2$ this is the $\pm 3\sqrt{R}$ region around the measurement realization (solid blue curve), which acts as an estimated mean in this case.

Fig. 5 shows the evolution of the estimated state pdf as per the UKF, for this particular Monte Carlo run, while Fig. 6 has the same data, but for the AGMUKF. The solid blue curve in each plot is the true state in the specified dimension – *i.e.* either error GRP w.r.t. the reference quaternion of the filter run, or plain angular velocity. The orange dashed curves enclose the 99.73% cumulative probability of the marginal pdf in that dimension.

For the UKF case, since the pdf is modelled as a Gaussian, for each state $i \in [1, n]$, this still is the $\pm 3\sigma_i$ offset from the filter estimated mean $\hat{x}_i$, where σ_i is the square root of the i-th diagonal element of P. Note that this applies to all prior and posterior estimations in all time steps – *i.e.* here sub-indices do not indicate the time step, as before, but the state dimension.

For the AGMUKF, however, at each time step of the filter (both prior and posterior), this region is defined as



(a) δp component $\boldsymbol{i}$

(b) $\omega^B_{B/I}$ component x

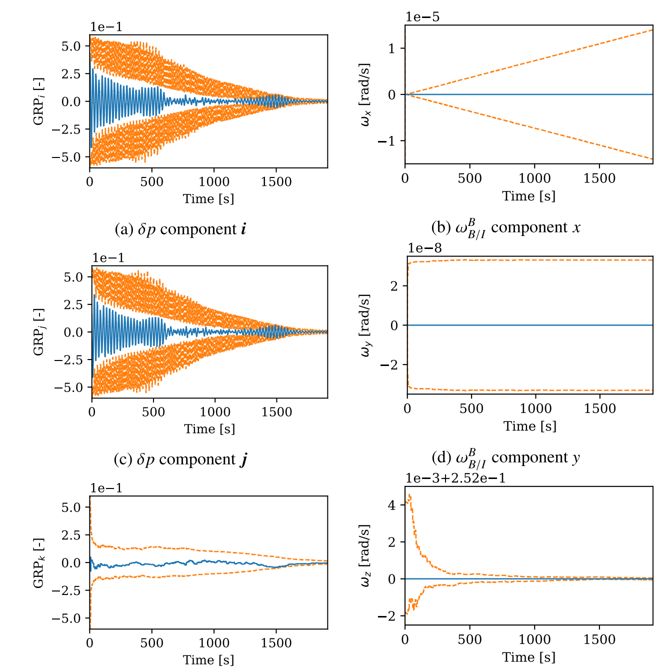
(c) δp component $\boldsymbol{j}$

(d) $\omega^B_{B/I}$ component y

(e) δp component $\boldsymbol{k}$

(f) $\omega^B_{B/I}$ component z

Fig. 5. One UKF run on the Wetterer and Jah (2009) scenario. Legend: truth (blue line); 3-σ envelope of the estimated pdf (dashed orange line).



(a) δp component $\boldsymbol{i}$

(b) $\omega^B_{B/I}$ component x

(c) δp component $\boldsymbol{j}$

(d) $\omega^B_{B/I}$ component y

(e) δp component $\boldsymbol{k}$

(f) $\omega^B_{B/I}$ component z

Fig. 6. One AGMUKF run on the Wetterer and Jah (2009) scenario. Legend: truth (blue line); 99.73% probability envelope of the estimated pdf (dashed orange line).

the sub-set Ω_i of the domain of each state such that, for probability $P_r = 99.73\%$,

$$
\begin{aligned}
&P_r = \int_{\Omega_i} \mathrm{pdf}_i(x_i)dx_i\,, \\
&\mathrm{pdf}_i(x_i) > \mathrm{pdf}_i(x_i'),\ \forall x_i \in \Omega_i \wedge x_i' \notin \Omega_i\,,
\end{aligned}
\tag{63}
$$

where

$$\mathrm{pdf}_i = \sum_{j=1}^{M} w^j \mathcal{N}\left(x_i | \hat{x}_i^j, \left(\sigma_i^j\right)^2\right) \tag{64}$$

is the univariate marginal GM of state dimension i. In words, this is the region with cumulative probability 99.73% such that the pdf at any point within this region evaluates higher than at any point without. It can be proven that Ω_i is unique given P_r and pdf_i.

The marginal pdfs estimated by each filter, overall, do not seem to be extremely different when comparing them within this particular Monte Carlo run. The AGMUKF estimates the pdf of the angular velocity as a bit more disperse, for the non rotating directions. Nonetheless, in both filters the truth stays within the 99.73% region for all states, even if it goes close to the border around the 1500s for the AGMUKF. Overall, both converge quite well by the end. Since this is a case tuned to work with the UKF version, it is to be expected that the AGMUKF reduces to almost a normal UKF.

Despite all this, the fact that each filter converges well once for a specific initial realization of the state and a specific realization of the light curve, does not guarantee that the filter will converge with all combinations. To assess this, the results of the 100 Monte Carlo runs need to be analysed as a whole, and not individually. This is done with the log likelihood of all the Monte Carlo runs (Schiemenz et al., 2020; Vittaldev and Russell, 2016)

$$LL = \sum_{i=1}^{MC} \left[\log \left(\sum_{j=1}^{M} w^j \mathcal{N}(x | \hat{x}^j, P^j) \right) \right]_i, \tag{65}$$

where the square brackets indicate each distinct Monte Carlo iteration i. In the UKF case, this reduces naturally to $M = 1$. In any circumstance, the higher the LL, the likelier it is that a filter estimates a consistent pdf of the state.

Fig. 7 shows the LL for the UKF; while Fig. 8, that of the AGMUKF. In both cases, the solid blue curve is the LL of the whole state, the orange dotted one is the LL applied only to the three attitude states, and the green dashed one is that of the angular velocity states.

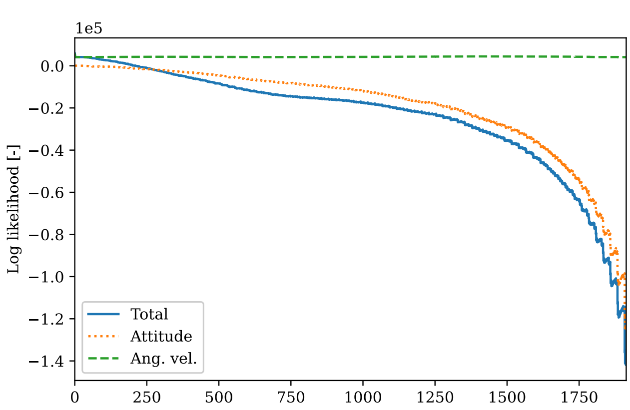


Fig. 7. Log likelihood averaged over 100 Monte Carlo UKF runs for the Wetterer and Jah (2009) scenario.

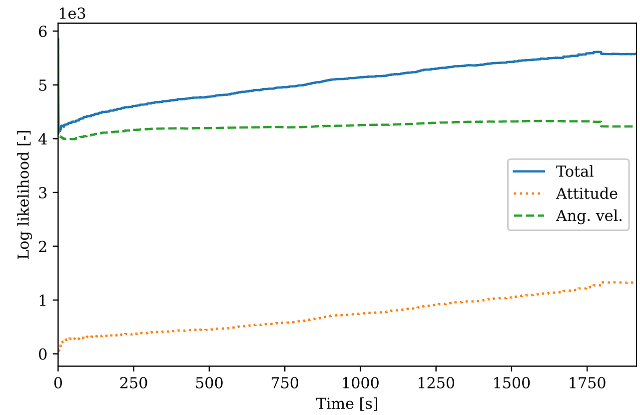


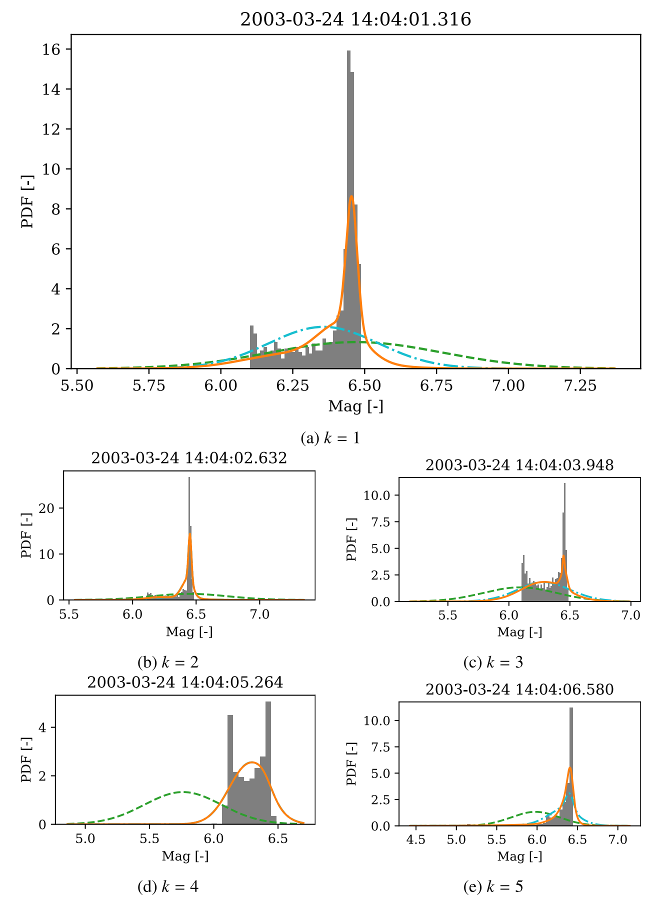
Fig. 8. Log likelihood averaged over 100 Monte Carlo AGMUKF runs for the Wetterer and Jah (2009) scenario.

These two plots do show a significant performance difference between the two filters. On the one hand, the log-likelihood of the UKF plummets as time goes by, which implies that the filter diverges in several of the runs. On the other hand, the LL of the AGMUKF steadily increases over time, indicating that it converges over all runs, consistently closing in around the true state as simulation time increases. This shows that the AGMUKF is clearly superior to the plain UKF in terms of stability, and that it is significantly likelier to estimate a consistent pdf.

The root cause for the superiority of the AGMUKF is likely the fact that it can model the pdf of the estimated measurement from the state with significantly higher accuracy. This is demonstrated by the results shown in Fig. 9, which is the equivalent of Fig. 2 but for the AGMUKF case. An additional curve is shown here – dash-dotted cyan – which represents the measurement pdf before the split. The solid orange one, then, is the measurement pdf that results after the split. When they coincide, it means that there is no need for splitting in that particular time step. By comparing Fig. 2 against Fig. 9, it is clear that the AGMUKF is able to model the measurement pdf, which is highly non Gaussian and frequently multimodal, with a higher degree of accuracy than the UKF.

Fig. 10 shows the history of the number of kernels for the AGMUKF run in question. The solid blue curve indicates mixture size over time, while the orange × marks appear whenever the mixture is refined, right after the split takes place. The first four fifths of the abscissa, up to the dash-dotted vertical line, is represented in a zoomed scale, to highlight the first 100s of simulation, when most of the splits happen. For the rest of the simulation time, expect at the very end, the kernel population gradually decreases as the kernels approach each other in KL-distance sense, which triggers merges once in a while.

Note, then, that even if at peak moments there are up to 100 kernels active, for the most part of the run the mixture is composed by less than 20. This demonstrates the effi-



(a) $k = 1$ (b) $k = 2$ (c) $k = 3$ (d) $k = 4$ (e) $k = 5$

Fig. 9. Probability distribution function of $\hat{z}_k$ for the first five measurements in the Wetterer and Jah (2009) scenario when using the AGMUKF. Legend: pdf of the observation (dashed green line); pdf of the measurement estimated from the state using UT before the split (dash-dotted cyan line), UT after the split (solid orange line) or Monte Carlo (gray histogram). Dates in the titles expressed in ISO format: YYYY-MM-DD hh:mm.ss.fff (f stands for millisecond).

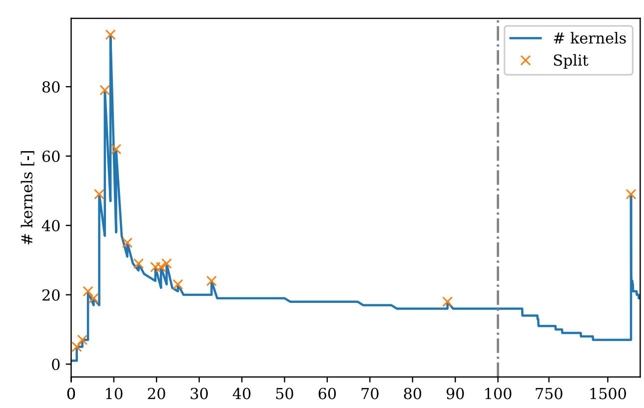


Fig. 10. Kernel number history for one AGMUKF run of the Wetterer and Jah (2009) scenario.

ciency of the AGMUKF architecture, which refines the mixture only when necessary, coarsening it as soon as possible to reduce the computational burden. Without parallelization, under a constant measurement frame rate, the AGMUKF uses approximately c times the computational resources of the UKF, where

$$c = \frac{1}{t_{N_t}} \int_0^{t_{N_t}} M(t)dt, \tag{66}$$

which is the area under the blue curve in Fig. 10. For the particular run presented here, this is $c \sim 11.7$. The AGMUKF has the addditional overhead of splitting and merging kernels, however. In particular, the computational needs of the merge step grow quadratically with M, since it needs to compute the relative distance between all kernels. Fortunately, both the merge step and all the kernel-by-kernel steps (*e.g.* propagation) can be parallelized, since the operations done on one kernel do not depend on the other kernels for most of the algorithm. Additionally, for each kernel, the UT propagation can be parallelized, too, since each sigma point can be processed independently from the others. Thus, overall, the execution time of both the UKF and the AGMUKF can be approximately divided by the number of available threads.

5.2. Highly non-linear case

The other test result studied in this paper is that of a highly non-linear case, where the initial uncertainty of the state is greater than the previous case. Keeping on with the comparison against existing literature of AD using different techniques, this subsection studies a scenario similar to the one presented in Linares et al. (2014a), Example 2. This scenario has been chosen for the similarity of the models used therein w.r.t. to the dynamic and measurement models of this paper, and because Linares et al. (2014a) use the PF for AD, the base of the current state-of-the art, and main competitor to the AGMUKF (see Section 1).

The main modification of the scenario used here w.r.t. Linares et al. (2014a) is that a different shape is used, since they do not provide information on the reflective properties of the surfaces of their test shape. To follow with the example from the previous section, the shape used is a 100-faceted cylinder approximation with 9m length and 3m diameter, with conic caps 1.5m long on either end, like the one in Fig. 3. The body x axis is parallel to the axial direction of the cylinder, while y and z lie on the orthogonal plane, forming a right-handed frame. The same reflective properties as in the previous section have been used. Regarding the filter, the same UKF- and AGMUKF-specific tunable parameters are kept from the previous section, too.

In the Monte Carlo run presented here, the filter is initialized with an initial diagonal covariance, with the δp components being 0.3932 and the ω ones, $2.350 \cdot 10^{-4}\mathrm{rad}^2/\mathrm{s}^2$, as in Linares et al. (2014a). The initial state is

$$\tilde{\boldsymbol{q}}_0 = 0.495 - 0.710\boldsymbol{i} + 0.293\boldsymbol{j} - 0.407\boldsymbol{k} \tag{67}$$

and

$$\hat{\omega}_0 = \begin{bmatrix} -2.50 \cdot 10^{-3} & 1.50 \cdot 10^{-3} & 1.31 \cdot 10^{-3} \end{bmatrix}^T \frac{\text{rad}}{\text{s}} . \tag{68}$$

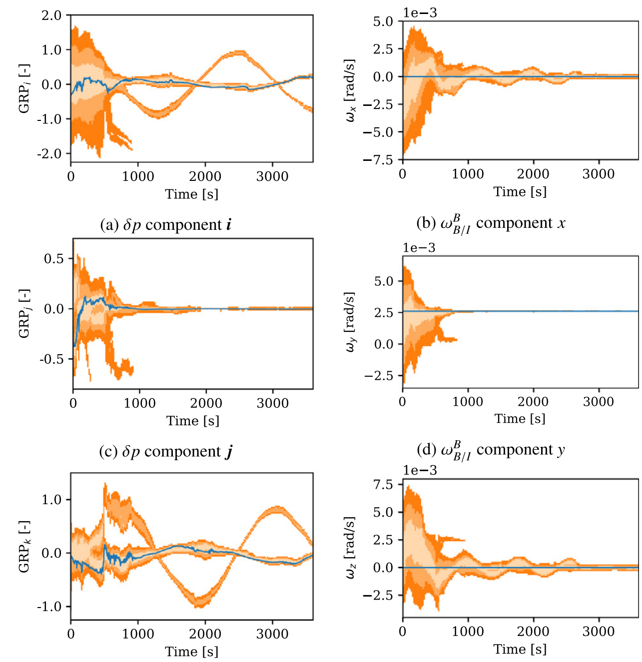
This results in the initial $\delta\hat{p}_0 = 0$ being at an Euclidean distance of ~ 1.13 from the true δp_0, in GRP space. Fig. 11 shows the measured light curve, where the zoomed plots include the bounds of the 3σ, or 99.73% probability region, with $R = 0.05^2$. The process noise used is

$$Q_k = \text{diag}\left(\begin{bmatrix} 0^{-12} & 10^{-12} & 10^{-12} & 10^{-14} & 10^{-14} & 10^{-14} \end{bmatrix}\right). \tag{69}$$

Again, net torque is 0.

Fig. 12 shows the evolution of the estimated pdf of this particular filter run. In contrast to Figs. 5 and 6, here the 1σ ($P_r = 68.27\%$), 2σ ($P_r = 95.45\%$) and 3σ ($P_r = 99.73\%$) probability regions are all represented in solid orange, from lighter to darker tone, instead of using bounds.

Within this run, the AGMUKF is able to constrain the state pdf well around the truth, as shown in Fig. 12. However, due to the high nonlinearity of the problem, coupled with the wide initial uncertainty in attitude, which almost gives no constrain to the initial state, the filter cannot fully decide among two different modes. These two modes have the same angular rate and $\boldsymbol{j}$ GRP component, but show a constant offset on the $\boldsymbol{i}$ and $\boldsymbol{k}$ GRP states (Figs. 12a and 12c). This means that there is a non-negligible probability that the measured light curve could be the result of either alternative attitude states, as far as the filter is concerned.



(a) δp component $\boldsymbol{i}$ (b) $\omega^B_{B/I}$ component x

(c) δp component $\boldsymbol{j}$ (d) $\omega^B_{B/I}$ component y

(e) δp component $\boldsymbol{k}$ (f) $\omega^B_{B/I}$ component z

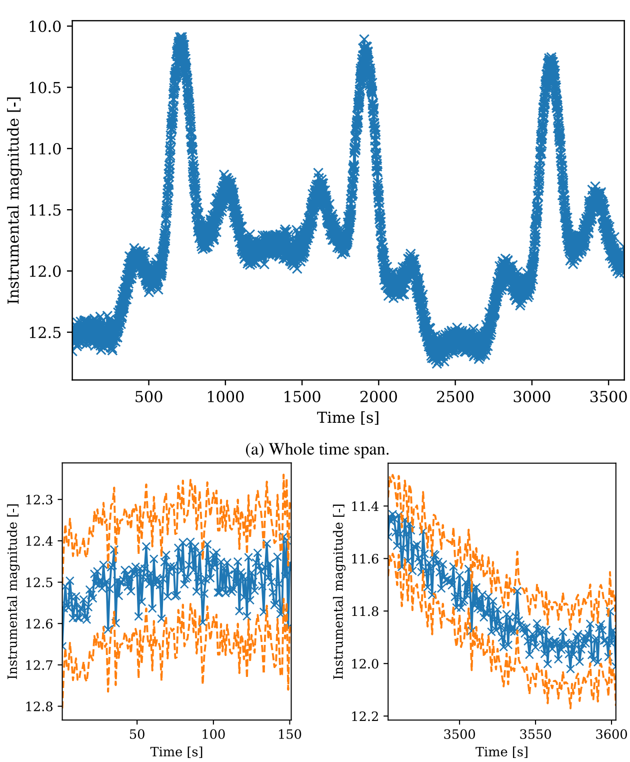
Fig. 12. One AGMUKF run on the Linares et al. (2014a) scenario. Legend: truth (blue line); 99.73% envelope of the estimated pdf (dark orange region), 95.45% envelope (middle orange region), 68.27% envelope (pale orange region).

This does not imply that the filter has failed to converge. In fact, the true state stays within the 1σ (palest orange) region through most of the simulation time. Thus, the mode that differs more from the truth is the one with a significantly lower share of the total probability. However, most importantly, the log likelihood of this single run increases throughout the simulation (Fig. 13), meaning that the filter tends to converge around the truth as new measurements com in.

This particular Monte Carlo run has been chosen in this paper precisely as a clear example of convergence around a multi-modal solution, a feature that is enabled by the use of Gaussian Mixtures.

This stability, however, is not preserved in a 100-samples Monte Carlo study, as in the previous scenario: while some runs have converged around a single attitude mode, or with multi-modal pdf, such as the one presented here, some other runs diverge, making the total LL plummet.

Since the initial greater covariance results in higher nonlinearity indices, both in the dynamic and the measurement models, but especially in the latter (see Fig. 14), the splitting step recommends significantly more splits, compared to the Wetterer and Jah (2009) scenario. This is clear in the plot from Fig. 15, where the first 700s are zoomed into show in more detail the region where most splits happen. The total number of kernels in a given time reaches the



(a) Whole time span.

(b) First 3 minutes. (c) Last 3 minutes.

Fig. 11. One noisy light curve of the Linares et al. (2014a) scenario. Blue crosses are brightness measurements, while the blue line is a visual guide. The orange dashed line is the 3-σ envelope of uncertainty around the light curve.

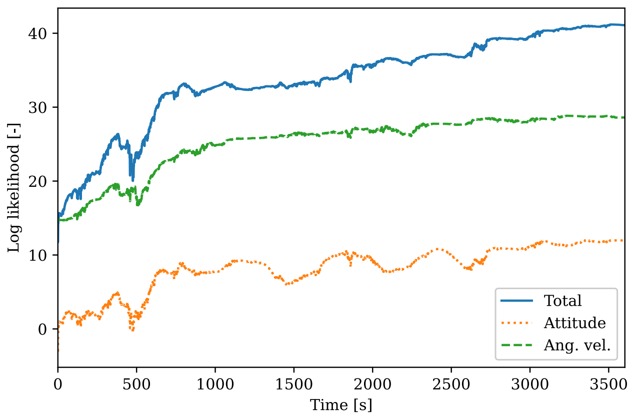


Fig. 13. Log likelihood averaged over a single Monte Carlo AGMUKF run for the Linares et al. (2014a) scenario.

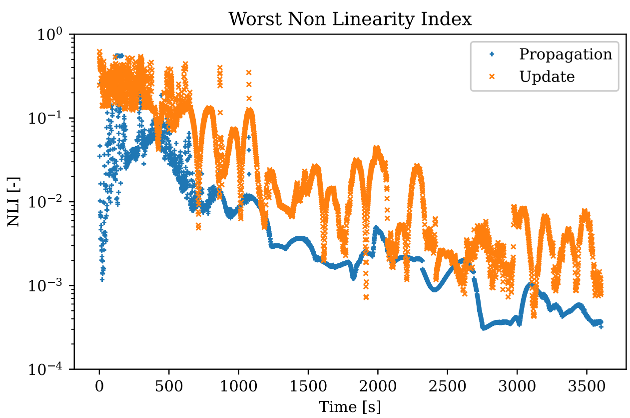


Fig. 14. Non-Linearity Index of one AGMUKF run for the Linares et al. (2014a) scenario.

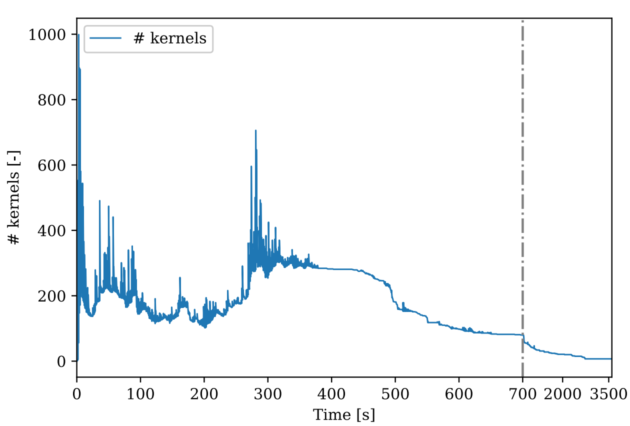


Fig. 15. Kernel number history for one AGMUKF run of the Linares et al. (2014a) scenario.

hard limit of 999 at the very beginning, suggesting that a higher bound might improve the results for this scenario in a general Monte Carlo sense.

Given that the 100-runs Monte Carlo study does not result in LL convergence, it is clear that several improvements could be attempted to stabilize the AGMUKF for this kind of higher initial uncertainty scenario. Some viable options would be to:

- increase the maximum allowed size of the population, as suggested above;
- make the splitting algorithm (Eq. 39) more aggressive by making it start splitting earlier (higher b) and at a faster rate (higher a) – this would result in a finer modelling of the measurement pdf;
- since the NLI is significantly high for the propagation step (Fig. 14), incorporate the split/merge strategy around it, too, as in Schiemenz et al. (2020) for the OD case; in this case, doing multiple prediction steps between measurements could help conserve a more faithful estimate of the state pdf;
- include better process noise tuning/modelling, as in Crassidis and Markley (2003), Coder et al. (2017);
- make the filter over-estimate the initial covariance, to ensure outlier initial conditions are still covered;
- initialize the filter with a GM of more than one kernel, with the goal to off-load the splitting algorithm with the (currently) inherent limit of 39 splits per direction per update of the Vittaldev and Russell (2016) library; this could be combined with the Expectation–Maximization (EM) algorithm, as in Yun and Zanetti (2020), to fit the initial mixture to some arbitrary initial pdf, such as the uniform distribution (Shuster, 2003);
- the 39-splits limit might also be palliated by recursively splitting plus consecutive merging – although this is not the optimal split solution, it might still benefit the filter by reducing the non-linearity of each kernel.

Computationally speaking, applying Eq. 66 to Fig. 15 yields an average number of kernels per iteration of roughly 70. Each kernel consists of an UKF composed by 13 particles (twice the number of error-states plus one). Thus, the results here have been obtained with an average of $13 \times 70 = 910$ particles. In contrast, the results from Linares et al. (2014a, Fig. 3) use roughly 20000 particles, one order of magnitude higher. Further tests are needed to assess how the AGMUKF and the PF perform in exactly identical scenarios, and how much of the computational overhead is taken by the AGMUKF splitting/merging part, versus the resampling and regularization tasks required by the PF. Nonetheless, these results show significant promise for the AGMUKF, and its potential efficiency benefits w.r.t. the PF.

6. Conclusions

This paper has introduced the process of non-linear sequential estimation aimed at the Attitude Determination (AD) of Resident Space Objects (RSOs) using light curves, when the shape and surface reflectance model is known. The dynamic and measurement models of the problem have been presented, and their significantly non-linear character has been emphasized.

An existing Non-Linearity Index (NLI) based on the Unscented Transform has been modified to allow comparison between different transformation models. This new NLI has been used within a Monte Carlo study of the measurement model to assess the inherent non-linearity of the problem. Results have shown that the measurement model is relatively less linear than the dynamic one. This has motivated the design of the Adaptive Gaussian Mixtures Unscented Kalman Filter (AGMUKF) for AD, based on a similar implementation of the AGMUKF for Orbit Determination (OD).

First, the convergence performance and stability of this new filter has been assessed against a scenario that had previously been proven to converge under the classical Unscented Kalman Filter (UKF) for AD. The results of a Monte Carlo study have shown that, although the convergence rate can be similar for particular runs, the AGMUKF is significantly likelier to always converge than the UKF.

Second, the AGMUKF has been tested against a more demanding scenario, for which the Particle Filter (PF), its main rival, had been previously proven to work. The AGMUKF has been shown to converge for an example run with significant initial uncertainty, successfully capturing the multimodal character of the state probability density function. Several suggestions have been given to potentially improve its stability against varied initial and noise conditions.

The results of this paper have introduced the novel AGMUKF for AD as potential alternative to PF-derived techniques, since it has been shown to converge in scenarios where the traditional PF does, too. In contrast to the PF, the AGMUKF manages the number of kernels (particles in PF lingo) more efficiently, since it can concurrently assess the non-linearity present in the model. Furthermore, the AGMUKF avoids the PF degeneracy problem, as well as its dependence on regularization. Finally, the AGMUKF naturally avoids the need to marginalize the PF when a part of the state evolves linearly through the involved models, as in Coder et al. (2017), because the NLI will naturally stay 0 in these directions.

The one caveat of the AGMUKF w.r.t. the PF is that the computational complexity of the merging step increases quadratically with the number of kernels. Significant improvement would be achieved by devising more efficient merging criteria.

Further work should include the application of the AGMUKF to other dynamic and measurement models, which *e.g.* target agile objects (Holzinger et al., 2012), to include shape- and reflectance-model uncertainties (Coder et al., 2017) or to fuse attitude and orbit estimation (Yun and Zanetti, 2020), or attitude and shape (Linares and Crassidis, 2018). Finally, the AGMUKF for AD should be tested against real-world data, as in Coder et al. (2018).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Quaternions

A.1. Quaternion algebra

The quaternion algebra, also known as Hamilton space and denoted here by $\mathbb{H}$, deals with quaternions, represented as

$$\boldsymbol{q} = q_r + q_i\boldsymbol{i} + q_j\boldsymbol{j} + q_k\boldsymbol{k} = [1 \quad \boldsymbol{i} \quad \boldsymbol{j} \quad \boldsymbol{k}]q, \tag{A.1}$$

where $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k} \in \mathbb{H}$ are the three independent hyper-imaginary parts, or basic quaternions, and $q = \begin{bmatrix} q_r & q_i & q_j & q_k \end{bmatrix}^T \in \mathbb{R}^4$ is the vector representation of $\boldsymbol{q}$. Much like complex numbers, quaternions have a real part, $\Re(\boldsymbol{q}) = q_r$, and a hyper-imaginary part, $\Im(\boldsymbol{q}) = \begin{bmatrix} q_i & q_j & q_k \end{bmatrix}^T$.

Addition is performed using the same rules for real numbers applied to the real and to each of the three imaginary parts,

$$\boldsymbol{q} + \boldsymbol{p} = (q_r + p_r) + (q_i + p_i)\boldsymbol{i} + (q_j + p_j)\boldsymbol{j} + (q_k + p_k)\boldsymbol{k}. \tag{A.2}$$

Multiplication is governed by the quaternion equation:

$$\boldsymbol{ii} = \boldsymbol{jj} = \boldsymbol{kk} = \boldsymbol{ijk} = -1. \tag{A.3}$$

One can multiply any two arbitrary quaternions by applying associative and distributive laws to each of the real and hyper-imaginary components. Note, however, that quaternion multiplication is not commutative.

The conjugate of a quaternion is obtained by reversing the sign of the hyper-complex part only:

$$\boldsymbol{q}^* = [1 \quad -\boldsymbol{i} \quad -\boldsymbol{j} \quad -\boldsymbol{k}]q. \tag{A.4}$$

With this, the absolute value of the quaternion can be obtained as

$$|\,\boldsymbol{q}\,| = \sqrt{\boldsymbol{q}\boldsymbol{q}^*} = \sqrt{q_r^2 + q_i^2 + q_j^2 + q_k^2}. \tag{A.5}$$

The inverse is such that

$$\boldsymbol{q}\boldsymbol{q}^{-1} = \boldsymbol{q}^{-1}\boldsymbol{q} = 1\,. \tag{A.6}$$

Using the above definitions, it is trivial to show that

$$\boldsymbol{q}^{-1} = \frac{\boldsymbol{q}^*}{|\,\boldsymbol{q}|^2}\,. \tag{A.7}$$

A.2. Attitude quaternion derivative

Given the quaternion that transforms from frame B at some time t to frame B^+ at some time $t + \Delta t$,

$$q_B^{B^+} = \left[\cos\left(\tfrac{\Delta\varphi}{2}\right) \quad e^T\sin\left(\tfrac{\Delta\varphi}{2}\right)\right]^T = \left[1 \quad e^T\tfrac{\Delta\varphi}{2}\right]^T + O\left(\Delta\varphi^2\right), \tag{A.8}$$

one can obtain the attitude at time $t + \Delta t$ from the one at time t by composing rotations (Eq. 9):

$$\boldsymbol{q}_{B^+}^I = \left(\boldsymbol{q}_B^{B^+}\right)^*\boldsymbol{q}_B^I\,. \tag{A.9}$$

One can take the limit to obtain the derivative function that governs the attitude quaternion:

$$\dot{\boldsymbol{q}}_B^I = \lim_{\Delta t\to 0}\frac{\left(\boldsymbol{q}_B^{B^+}\right)^*\boldsymbol{q}_B^I - \boldsymbol{q}_B^I}{\Delta t} = \lim_{\Delta t\to 0}\frac{\cancel{(1-1)\boldsymbol{q}_B^I} - \left(\boldsymbol{e}\frac{\Delta\varphi}{2}\right)\boldsymbol{q}_B^I + (1+i+j+k)O\left(\Delta\varphi^2\right)}{\Delta t}\,. \tag{A.10}$$

By defining

$$\omega_{B/I}^B := \lim_{\Delta t\to 0}\frac{e\Delta\varphi}{\Delta t}\,, \tag{A.11}$$

one can resolve the above limit to

$$\dot{\boldsymbol{q}}_B^I = -\frac{1}{2}\omega_{B/I}^B\boldsymbol{q}_B^I + \cancel{\lim_{\Delta t\to 0}(1+i+j+k)O\left(\Delta t^2\|\omega_{B/I}^B\|^2\right)}\,. \tag{A.12}$$

Thus, $\omega_{B/I}^B$ can be read as the *angular velocity of frame B relative to frame I, expressed in B coordinates.*

Appendix B. Generalized rodriges parameters derivations

B.1. Taylor expansion of the GRP

Given

$$p_{a,f} = f\frac{\mathrm{q}}{a + q_r}\,, \tag{B.1}$$

Taylor expansion around φ (see Eq. 13) yields

$$p_{a,f} = f\frac{e\varphi}{2(a+1)} + O\left(|\varphi|^3\right). \tag{B.2}$$

Thus, if $f = 2(a+1)$, it cancels with the denominator and makes p equal to $e\varphi$ up to second order approximation.

B.2. Change of the reference quaternion

Let us define a reference attitude quaternion $\tilde{\boldsymbol{q}}$ and two other quaternions

$$\boldsymbol{q} = \delta\boldsymbol{q}^*\tilde{\boldsymbol{q}} \tag{B.3}$$

and

$$\tilde{\boldsymbol{q}}' = (\delta\tilde{\boldsymbol{q}}')^*\tilde{\boldsymbol{q}} \tag{B.4}$$

whose GRP error representations are, then, $\delta p = p(\delta\boldsymbol{q})$ and $\delta\tilde{p}' = p(\delta\tilde{\boldsymbol{q}}')$. The goal is to find $\delta p'$ that represents the same quaternion $\boldsymbol{q}$, but using $\tilde{\boldsymbol{q}}'$ as reference quaternion. This is, we need to find $\delta p' = p(\delta\boldsymbol{q}')$ such that

$$\delta\boldsymbol{q}^*\tilde{\boldsymbol{q}} = (\delta\boldsymbol{q}')^*\,\tilde{\boldsymbol{q}}' \Rightarrow \delta\boldsymbol{q}^*\cancel{\tilde{\boldsymbol{q}}} = (\delta\boldsymbol{q}')^*\,(\delta\tilde{\boldsymbol{q}}')^*\,\cancel{\tilde{\boldsymbol{q}}}\,, \tag{B.5}$$

which yields

$$\delta\boldsymbol{q}' = (\delta\tilde{\boldsymbol{q}}')^*\delta\boldsymbol{q} \Rightarrow \delta p' = p\left(\left(p^{-1}(\delta\tilde{p}')\right)^*p^{-1}(\delta p)\right). \tag{B.6}$$

This expression can be expanded within first order as

$$\begin{aligned}\delta p'(\delta\tilde{p}', \delta p) = \delta p'(0,0) &+ \left.\frac{\partial\delta p'}{\partial\delta\tilde{p}'}\right|_{0,0}(\delta\tilde{p}' - 0)\\ &+ \left.\frac{\partial\delta p'}{\partial\delta p}\right|_{0,0}(\delta p - 0)\\ &+ O\left(\|\delta\tilde{p}'\|^2 + \|\delta p\|^2\right).\end{aligned} \tag{B.7}$$

The Jacobians involved can be obtained by applying the derivative chain rule, which yields

$$\frac{\partial\delta p'}{\partial\delta\tilde{p}'} = \frac{\partial\delta p'}{\partial\delta q'}\frac{\partial\delta q'}{\partial\delta\tilde{q}'}\frac{\partial\delta\tilde{q}'}{\partial\delta\tilde{p}'} \tag{B.8}$$

and

$$\frac{\partial\delta p'}{\partial\delta p} = \frac{\partial\delta p'}{\partial\delta q'}\frac{\partial\delta q'}{\partial\delta q}\frac{\partial\delta q}{\partial\delta p}\,. \tag{B.9}$$

Starting with the first term in both derivatives, which is the same, we can compute it by differentiating Eq. B.1 with respect to q, as a function of q, which yields

$$\frac{\partial p}{\partial q}(q) = \frac{1}{a + q_r}\left[-p(q) \quad fI_3\right], \tag{B.10}$$

where I_i is the identity matrix in $\mathbb{R}^{i\times i}$. The last term can likewise be obtained by finding the inverse derivative, by differentiating Eq. 18:

$$\frac{\partial q}{\partial p}(p) = \begin{bmatrix} g(p) \\ \frac{1}{f}pg(p) + \frac{a + q_r(p)}{f}I_3 \end{bmatrix}, \tag{B.11}$$

where

$$\begin{aligned} g(p) = \frac{\partial q_r}{\partial p}(p) = \Bigg(&\frac{-2a + f(1-a^2)\left(f^2 + (1-a^2)\|p\|^2\right)^{-\frac{1}{2}}}{f^2 + \|p\|^2}\\ &-2\frac{-a\|p\|^2 + f\sqrt{f^2 + (1-a^2)\|p\|^2}}{\left(f^2 + \|p\|^2\right)^2}\Bigg)p^T\,. \end{aligned} \tag{B.12}$$

Finally, the middle terms are

$$\frac{\partial\delta q'}{\partial\delta\tilde{q}'} = \begin{bmatrix}\delta q_r & \delta\mathrm{q}^T\\ \delta\mathrm{q} & -\delta q_r I_3 + [\delta\mathrm{q}]_\times\end{bmatrix} \tag{B.13}$$

and

$$\frac{\partial \delta q'}{\partial \delta q} = \begin{bmatrix} \delta \tilde{q}_r & \delta \tilde{\mathfrak{q}}^T \\ -\delta \tilde{\mathfrak{q}} & \delta \tilde{q}_r I_3 - [\delta \tilde{\mathfrak{q}}]_\times \end{bmatrix}, \tag{B.14}$$

which come from differentiating the multiplication rules defined by Eq. A.3, applied to the vector form of $\delta \boldsymbol{q}'$ as defined in Eq. B.6. The operator $[\cdot]_\times : \mathbb{R}^3 \to \mathbb{R}^{3\times 3}$ takes the cross product matrix, such that $[v]_\times u = v \times u$, for any $u, v \in \mathbb{R}^3$.

Evaluating these derivatives at $\delta \tilde{p}' = 0$ and $\delta p = 0$ and substituting in Eq. B.7 results in

$$\delta p'(0,0) = 0\,, \tag{B.15a}$$

$$\left.\frac{\partial \delta p'}{\partial \delta \tilde{p}'}\right|_{0,0} = -I_3 \text{and} \tag{B.15b}$$

$$\left.\frac{\partial \delta p'}{\partial \delta p}\right|_{0,0} = I_3\,; \tag{B.15c}$$

thus

$$\delta p' = \delta p - \delta \tilde{p}' + O\left(\|\delta \tilde{p}'\|^2 + \|\delta p\|^2\right). \tag{B.16}$$

Note that this result is reached regardless of the values of a and f.

Appendix C. The unscented transform

This section provides a compact reproduction of the Unscented Transform (UT) equations, for the reader's reference (Julier and Uhlmann, 2004).

Let us define a random variable $X \in \mathbb{R}^n$ which follows a statistical distribution whose first (mean) and second (covariance) moments are μ_x and P_x, respectively. Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a map that applies a non-linear transformation to X, such that a new random variable $Y \in \mathbb{R}^m$ can be obtained as

$$Y = f(X) + V\,, \tag{C.1}$$

where $V \in \mathbb{R}^m$ is the random variable that represents the uncertainty associated to the physical model represented by f, a.k.a as model noise. Its associated distribution has 0 mean and covariance P_v.

The goal of the UT is to obtain the mean and covariance of the distribution associated to Y. It uses $\alpha, \beta, \kappa \in \mathbb{R}$ as tunable parameters.

First, $2n+1$ sigma points are computed with

$$\mathcal{X}(0) = \mu_x\,, \tag{C.2a}$$

$$\mathcal{X}(i) = \mathcal{X}(0) + \sqrt{n+\lambda} S_i \text{and} \tag{C.2b}$$

$$\mathcal{X}(i+n) = \mathcal{X}(0) - \sqrt{n+\lambda} S_i\,, \tag{C.2c}$$

for all $i \in [1, n]$, where S_i is the i-th column of the Cholesky decomposition of $P_x = SS^T$ and

$$\lambda = \alpha^2(n+\kappa) - n\,. \tag{C.3}$$

Then, the sigma cloud is transformed under the map f, obtaining the transformed sigma cloud

$$\mathcal{Y}(i) = f(\mathcal{X}(i)), \forall i \in [0, 2n]\,. \tag{C.4}$$

Finally, the mean of the transformed distribution is estimated as

$$\hat{\mu}_y = \sum_{i=0}^{2n} w_m(i) \mathcal{Y}(i) \tag{C.5}$$

and its covariance as

$$P_y = \sum_{i=0}^{2n} w_c(i) \left(\mathcal{Y}(i) - \hat{\mu}_y\right)\left(\mathcal{Y}(i) - \hat{\mu}_y\right)^T + P_v\,. \tag{C.6}$$

The weights w_m and w_c are obtained as

$$w_m(0) = \frac{\lambda}{n+\lambda}, \, w_c(0) = w_m(0) + 1 - \alpha^2 + \beta, w_m(i) = w_c(i) = \frac{1}{2(n+\lambda)}\,, \tag{C.7}$$

for all $i \in [1, 2n]$.

Likewise, the cross covariance of X and Y may be estimated using the UT as

$$cov(X, Y) = \sum_{i=0}^{2n} w_c(i) (\mathcal{X}(i) - \mu_x) \left(\mathcal{Y}(i) - \hat{\mu}_y\right)^T. \tag{C.8}$$

The UT preserves the first and second statistical moments of any transformed distribution. Additionally, proper tuning of the β parameter can guarantee the third and fourth moments when dealing with Gaussian distributions. The parameters α and κ control the spread of the sigma points.

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